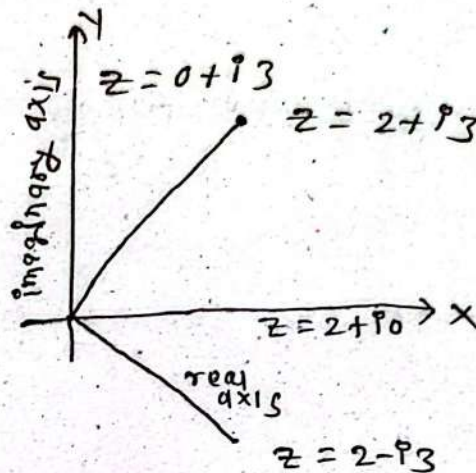


Chapter - 1

Complex Analysis

Complex numbers :-

A number of the form $a+ib$ where a, b are real numbers and $i = \sqrt{-1}$ is called complex number and denoted by z i.e. $z = a+ib$ or $z = x+iy$.



A pair of complex number $x+iy$ and $x-iy$ are said to be conjugate of each other and conjugate of $z = x+iy$ is $x-iy$ denoted by $\bar{z}$ i.e. $\bar{z} = x-iy$.

The modulus or absolute value of complex number $z = x+iy$ is denoted by $|z|$ and defined by
$$|z| = \sqrt{x^2+y^2}$$

Polar Form of complex numbers

By changing the cartesian co-ordinate (x, y) to polar form (r, θ) we replace $x = r \cos \theta$, $y = r \sin \theta$.

Then, the complex number $z = x+iy$

$$\text{or, } z = r \cos \theta + i r \sin \theta.$$

$$\text{or, } z = r (\cos \theta + i \sin \theta)$$

$$\therefore \boxed{z = r e^{i\theta}}$$

power of a complex number

Let $z = re^{i\theta}$ be non-zero complex number, the integral power of the complex number is given by

$$z^n = (re^{i\theta})^n$$

$$= r^n e^{in\theta}$$

$$z^n = r^n [\cos n\theta + i \sin n\theta]$$

①

$$\frac{3+4i}{7+i8}$$

[Express the following complex number in $x+iy$]

Soln

$$\text{Let } x+iy = \frac{3+4i}{7+i8} \times \frac{7-i8}{7-i8}$$

$$= \frac{21 - 24i + 28i - 32i^2}{49 - i^2 8^2}$$

$$\text{or, } x+iy = \frac{53}{113} + \frac{i4}{113}$$

Equation real & imaginary part

$$\text{i.e. } x = \frac{53}{113} \text{ \& } y = \frac{4}{113}$$

* Complex Function

Let $z = x+iy$ and $w = u(x,y) + iv(x,y)$ be two complex number in a certain region R of complex plane. Then a function defined on R is a rule that assigns to every z in R a complex number w called complex function of f and z that is $f(z) = w$

$$\text{i.e. } f(z) = u(x,y) + iv(x,y)$$

Analytic Function

⇒ A single-valued function of the complex variable is analytic or regular holomorphic in a certain domain. If it has a derivative of each point in the domain.

If the function $f(z)$ is differentiable at a single point, then it is not said to be analytic function.

← The necessary condition for $f(z)$ to be analytic (The Cauchy - Riemann equation)

$$1. f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

$$2. \frac{\partial u}{\partial x}(x,y) = u_x = \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x, y) - u(x, y)}{\Delta x}$$

$$3. \frac{\partial u}{\partial y}(x,y) = u_y = \lim_{\Delta y \rightarrow 0} \frac{u(x, y + \Delta y) - u(x, y)}{\Delta y}$$

YVT state and prove Cauchy - Riemann equation.

- If a function $f(z) = u(x,y) + iv(x,y)$ is differentiable at any point $z = x + iy$, then the first order partial derivatives u_x, u_y, v_x, v_y must exist and satisfy

$$\cancel{u_x = v_y} \quad \& \quad \cancel{u_y = -v_x}$$
$$u_x = v_y \quad \& \quad u_y = -v_x$$

proof:

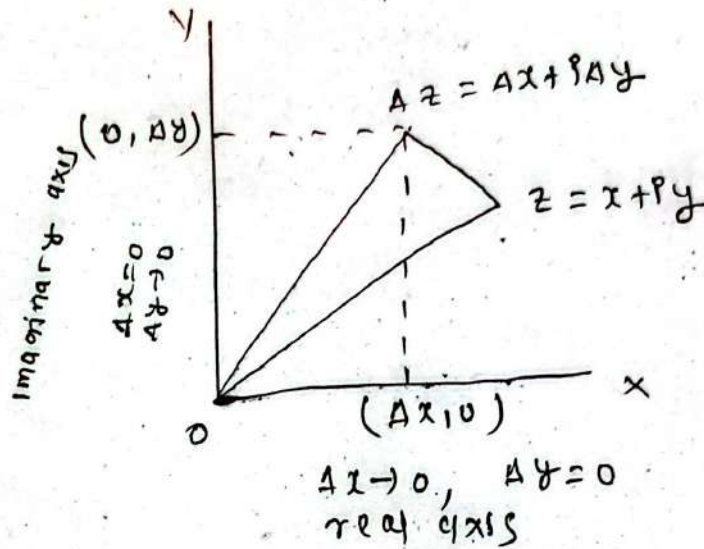
Since $f(z) = u(x,y) + iv(x,y)$ is differentiable at a point $z = x + iy$

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} \quad \text{provided the limit exists}$$

so, let $\Delta z = \Delta x + i\Delta y$

$$\text{or, } f'(z) = \lim_{(\Delta x + i\Delta y) \rightarrow 0} \left[\frac{u(x+\Delta x, y+\Delta y) + iv(x+\Delta x, y+\Delta y) - u(x, y) - iv(x, y)}{\Delta x + i\Delta y} \right]$$

$$\text{or, } f'(z) = \lim_{(\Delta x + i\Delta y) \rightarrow 0} \left[\frac{u(x+\Delta x, y+\Delta y) - u(x, y)}{\Delta x + i\Delta y} + i \left(\frac{v(x+\Delta x, y+\Delta y) - v(x, y)}{\Delta x + i\Delta y} \right) \right] \quad \text{--- (1)}$$



Taking Δz to be wholly real so, that $\Delta y = 0$, $\Delta x \rightarrow 0$ in eqn (1)

$$f'(z) = \lim_{\Delta x \rightarrow 0} \left[\frac{u(x+\Delta x, y) - u(x, y)}{\Delta x} + i \left\{ \frac{v(x+\Delta x, y) - v(x, y)}{\Delta x} \right\} \right]$$

$$f'(z) = u_x + i v_x \quad \text{--- (2)}$$

and taking Δz to be wholly imaginary so, that $\Delta x = 0$, $\Delta y \rightarrow 0$ in eqn- (1)

$$f'(z) = \lim_{\Delta y \rightarrow 0} \left[\frac{u(x, y+\Delta y) - u(x, y)}{i\Delta y} + i \left\{ \frac{v(x, y+\Delta y) - v(x, y)}{i\Delta y} \right\} \right]$$

$$\text{or, } f'(z) = \frac{1}{i} u_y + v_y$$

$$\text{or, } f'(z) = v_y - i u_y \quad \text{--- (3)}$$

Now, From eqn (2) & (3) we get.

$$u_x + i v_x = v_y + i(-u_y)$$

Equation real & imaginary parts,

$$u_x = v_y \quad \& \quad v_x = -u_y$$

$$u_y = -v_x$$

Necessary condition

If $f(z)$ is analytic satisfy C-R eqn.

$$u_x = v_y \quad \& \quad u_y = -v_x$$

Sufficient condition for $f(z)$ to be analytic

If $u_x = v_y$ & $v_x = -u_y$ then $f(z)$ is analytic

1. Show that $f(z) = z^2$ is analytic.

Soln

here $f(z) = z^2$

$$\text{let } u + i v = (x + i y)^2 \\ = x^2 + i 2xy - y^2$$

$$\text{or, } u + i v = (x^2 - y^2) + i 2xy$$

Equating real & imaginary parts we get,

$$u = x^2 - y^2, \quad u_x = 2x, \quad u_y = -2y$$

$$v = 2xy, \quad v_x = 2y, \quad v_y = 2x$$

Now, $u_x = 2x$ & $u_y = -2y$

$$= v_y$$

$$\therefore v_x = -v_y$$

$$\therefore u_x = v_y$$

hence $f(z) = z^2$ is analytic

2. $F(z) = z^3$

Let $u+iv = (x+iy)^3$

or, $u+iv = x^3 + (iy)^3 + 3x^2iy + 3x(iy)^2 + (iy)^3$

or, $u+iv = x^3 - 3xy^2 + 3x^2iy - 3xy^2 - y^3$

or, $u+iv = (x^3 - 3xy^2) + i(3x^2y - y^3)$

Now, comparing real and imaginary parts we get

$u = x^3 - 3xy^2, \quad u_x = 3x^2 - 3y^2, \quad u_y = -6xy$

$v = 3x^2y - y^3, \quad v_x = 6xy, \quad v_y = 3x^2 - 3y^2$

Now, $u_x = 3x^2 - 3y^2 = v_y$

$u_y = -6xy = v_x$

Hence $F(z) = z^3$ is analytic.

C-R eqn in cartesian form

$u_x = v_y$ & $u_y = -v_x$ $\frac{\partial}{\partial x}(F(x^2)) = F(x^2) + 2x$

polar form of C-R eqn

we have C-R eqn in cartesian form is

$u_x = v_y$ & $u_y = -v_x$

also, we know that,

$f(z) = u+iv$, putting $z = re^{i\theta}$

$f(re^{i\theta}) = u(r, \theta) + iv(r, \theta) \quad \text{--- (1)}$

Differentiating eqn (1) partially w.r. to 'r'

$f'(re^{i\theta}) e^{i\theta} = \frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} \quad \text{--- (2)}$

$f'(re^{i\theta}) \cdot i r e^{i\theta} = \frac{\partial u}{\partial \theta} + i \frac{\partial v}{\partial \theta}$

$$f'(re^{i\theta}) e^{i\theta} = \frac{1}{ir} \frac{\partial v}{\partial \theta} + \frac{i}{ir} \frac{\partial v}{\partial \theta} \quad (3)$$

From eqn (2) & (3)

$$\frac{\partial v}{\partial r} + \frac{r \partial v}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta} - i \frac{1}{r} \frac{\partial v}{\partial \theta}$$

Equating real & imaginary parts,

$$\frac{\partial v}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta} \quad \frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial v}{\partial \theta}$$

$$\frac{\partial v}{\partial \theta} = -r \frac{\partial v}{\partial r}$$

Harmonic Function

A real valued function $u(x, y)$ of two variables x and y is said to be harmonic function in a certain domain in xy -plane if it has continuous partial derivative of the 1st & 2nd order and satisfy Laplace eqn.

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Q Show that the function $u(x, y) = 3x^2y + x^2 - y^3 - y^2$ is a harmonic function.

Q Find the corresponding harmonic conjugate $v(x, y)$ of $u(x, y)$ such that $f(z) = u + iv$ is an analytic function.

Soln.

$$u(x, y) = 3x^2y + x^2 - y^3 - y^2$$

$$\frac{\partial u}{\partial x} = 6xy + 2x \quad , \quad \frac{\partial u}{\partial y} = 3x^2 - 3y^2 - 2y$$

$$\frac{\partial^2 u}{\partial x^2} = 6y + 2 \quad , \quad \frac{\partial^2 u}{\partial y^2} = -6y - 2$$

Now,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 6y + 2 - 6y - 2$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{hence } u \text{ is a harmonic function.}$$

For corresponding harmonic conjugate, since $f(z)$ is analytic, by C-R equation,

$$u_x = v_y \text{ \& } u_y = -v_x$$

$$v_y = 6xy + 2x$$

Integrating it partially w.r. to 'y'

$$v = 6x \frac{y^2}{2} + 2xy + \phi(x)$$

$$\text{or, } v = 3xy^2 + 2xy + \phi(x) \quad \text{--- (1)}$$

diff. (1) partially w.r. to 'x'

$$v_x = 3y^2 + 2y + \phi'(x)$$

By C-R eqn.

$$v_x = -u_y$$

$$\text{Now, } -4y = 3y^2 + 2y + \phi'(x)$$

$$\text{or, } -(3x^2 - 3y^2 - 2y) = 3y^2 + 2y + \phi'(x)$$

$$\text{or, } -3x^2 + 3y^2 + 2y = 3y^2 + 2y + \phi'(x)$$

$$\text{or, } \phi'(x) = -3x^2$$

$$\therefore \phi'(x) = -3x^2$$

Integrating it wr to x

$$\phi(x) = \frac{-3x^2}{3} + C$$

$$\phi(x) = -x^3 + C$$

From (1)

$$v(x, y) = 3xy^2 + 2xy - x^3 + C$$

For analytic function.

$$f(z) = u + iv$$

$$= 3x^2y + x^2 - y^2 - y^3 + i(3xy^2 + 2xy - x^3 + C)$$

Qn. 2

$$y = \frac{x}{x^2 + y^2}$$

soln.

$$d\left(\frac{y}{x}\right) = \frac{y y' - y' y}{y^2}$$

$$\frac{d}{dx} (x^2 + y^2)^2 = 2(x^2 + y^2) 2x$$

$$y (x+y) = \frac{x}{x^2 + y^2}$$

$$4x = \frac{(x^2 + y^2) \cdot 1 - x(2x + 0)}{(x^2 + y^2)^2}$$

$$4x = \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

Again diff.

$$4_{xx} = \frac{(x^2 + y^2)^2 (-2x) - (y^2 - x^2) 2(x^2 + y^2) \cdot 2x}{(x^2 + y^2)^4}$$

$$4_{xx} = \frac{2x(x^2 + y^2) [-(x^2 + y^2) - 2(y^2 - x^2)]}{(x^2 + y^2)^4}$$

$$4_{xx} = \frac{2x(x^2 - 3y^2)}{(x^2 + y^2)^3}$$

diff ① partially w.r to y.

$$4_y = \frac{(x^2 + y^2) \cdot 0 - x(0 + 2y)}{(x^2 + y^2)^2}$$

$$4_y = \frac{-2xy}{(x^2 + y^2)^2}$$

Again diff.

$$4_{yy} = \frac{(x^2 + y^2)^2 (-2x) - (-2xy) 2(x^2 + y^2) \cdot 2y}{(x^2 + y^2)^4}$$

$$4_{yy} = \frac{-2x(x^2 + y^2) [x^2 + y^2 - 4y^2]}{(x^2 + y^2)^4}$$

$$u_{yy} = \frac{-2x [x^2 - 3y^2]}{(x^2 + y^2)^3}$$

Now,

$$\frac{\partial^2 u}{\partial^2 x} + \frac{\partial^2 u}{\partial^2 y} = 0$$

hence u is a harmonic function.
For corresponding harmonic conjugate v of u
By C-R eqn.

$$u_x = v_y \quad \& \quad -u_y = v_x$$

$$v_x = -u_y$$

$$v_x = \frac{-(-2xy)}{(x^2 + y^2)^2}$$

$$\text{or, } v_x = \frac{2xy}{(x^2 + y^2)^2}$$

$$\int \frac{2x}{x^2 + y^2} dx = \log(x^2 + y^2)$$

$$\int \frac{2x}{(x^2 + y^2)^2} dx = \frac{(x^2 + y^2)^{-2+1}}{-2+1}$$

Integrating w.r to x

$$v = \frac{y(x^2 + y^2)^{-2+1}}{-1} + \phi(y)$$

$$v = \frac{-y}{(x^2 + y^2)} + \phi(y) \quad \dots (2)$$

diff w.r to y

$$v_y = \frac{(x^2 + y^2)(-1) - (-y)2y}{(x^2 + y^2)^2} + \phi'(y)$$

$$\text{or, } v_y = \frac{-x^2 - y^2 + 2y^2}{(x^2 + y^2)^2} + \phi'(y)$$

$$v_y = \frac{y^2 - x^2}{(x^2 + y^2)^2} + \phi'(y)$$

By C-R $u_x = v_y$

$$u_x = \frac{y^2 - x^2}{(x^2 + y^2)^2} + \phi'(y)$$

$$\frac{y^2 - x^2}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2} + \phi'(y)$$

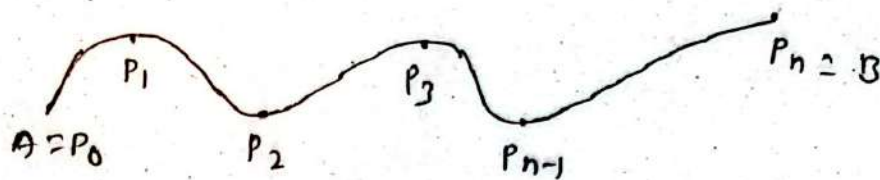
$\phi(y) = 0$ Integrating w.r to y

$$\phi(y) = C$$

From (2)

$$v = \frac{-y}{x^2 + y^2} + C$$

Complex Integration



Let $f(z)$ be a continuous function of complex variables $z = x + iy$ defined at all points of a curve C having end point A & B we sub-divide the curve C into n parts at the points.

$$A = P_0(z_0) + P_1(z_1) + P_2(z_2) - \dots - P_n(z_n) = B$$

let $\int z_i = z_i - z_{i-1}$ and ξ_i be any point on the arc $P_{i-1} P_i$ of the limit of a sum,

$$\sum_{i=1}^n f(\xi_i) \int z_i \text{ exists as } n \text{ tends to infinity.}$$

i.e. the chord $\int z_i$ approaches to zero then we can write $\int_C f(z) dz = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i) \int z_i$ or

$$\text{writing } f(z) = u + iv \text{ \& } dz = dx + i dy$$

$$\begin{aligned} \text{Then } \int_C f(z) dz &= \int_C (u + iv) (dx + i dy) \\ &= \int_C (u dx - \end{aligned}$$

Q

Evaluate $\int_C \bar{z} dz$, where C is the upper half of the circle $|z|=1$ joining

$$\int_C \bar{z} dz$$

$$z = re^{i\theta}$$

$$\therefore r=1$$

$$z = e^{i\theta}$$

$$\frac{dz}{d\theta} = e^{i\theta} \times i$$

$$dz = i e^{i\theta} d\theta$$

$$\bar{z} = e^{-i\theta}$$

$$= \int_0^\pi e^{-i\theta} \cdot i e^{i\theta} d\theta$$

$$= \int_0^\pi i d\theta = i \int_0^\pi d\theta$$

$$= i [\theta]_0^\pi$$

$$= i [\pi - 0]$$

$$= i\pi$$

Ans

Evaluate $\int_C (\bar{z} + 2z) dz$ along the circle $x^2 + y^2 = 1$

Soln

$$\int_C (\bar{z} + 2z) dz$$

$$x^2 + y^2 = 1 \quad \text{--- (1)}$$

comparing eqn (1) with

$$x^2 + y^2 = r^2$$

$$r^2 = 1$$

$$r = 1$$

$$z = re^{i\theta}$$

$$z = e^{i\theta}$$

$$\frac{dz}{d\theta} = e^{i\theta} \times i$$

$$dz = ie^{i\theta} d\theta$$

$$\bar{z} = e^{-i\theta}$$

$$= \int_C (\bar{z} + 2z) dz$$

$$= \int_0^{2\pi} (e^{-i\theta} + 2e^{i\theta}) ie^{i\theta} d\theta$$

$$= \int_0^{2\pi} [i + 2ie^{2i\theta}] d\theta$$

$$= \int_0^{2\pi} \left[i + 2 \left[i\theta + 2ie^{2i\theta} \times \frac{1}{2i} \right] \right] d\theta$$

$$= \left[i\theta + e^{2i\theta} \right]_0^{2\pi}$$

$$= \left[(i2\pi + e^{2i(2\pi)}) \right] - \left[(i(0) + e^0) \right]$$

$$= \left[2\pi i + e^{4\pi i} - 1 \right]$$

Note:- यदि $z = -i$ to $z = i$ है तो $\arg + 3\pi/2$ to $\pi/2$ है।

Q Evaluate $\int_C |z| dz$ where C is the left half of unit circle $|z| = 1$ From $z = -i$ to $z = i$

Soln

$$\int_C |z| dz$$

$$\int_C dz$$

$$\therefore |z| = r = 1$$

$$z = re^{i\theta}$$

$$z = e^{i\theta}$$

$$dz = e^{i\theta} \times i d\theta$$

$$= \int_{3\pi/2}^{\pi/2} 1 \cdot e^{i\theta} d\theta$$

$$= \left[1 \cdot e^{i\theta} \times \frac{1}{i} \right]_{3\pi/2}^{\pi/2}$$

$$= e^{i\pi/2} - e^{i3\pi/2}$$

Ans

Further calculation.

$$= (\cos \pi/2 + i \sin \pi/2) - (\cos 3\pi/2 + i \sin 3\pi/2)$$

$$= (0 + i) - (0 + i(-1))$$

$$= (i) - (-i)$$

$$= i + i$$

$$= 2i$$

Ans

Show that $\int_C \log z \, dz = 2\pi i$ where C is the unit circle in the z -plane.

soln

$$\int_C \log z \, dz$$

$$z = r e^{i\theta}$$

$$z = e^{i\theta}$$

$$dz = i e^{i\theta} d\theta$$

$$\log z = \log e^{i\theta} = i\theta$$

$$= \int_0^{2\pi} i\theta \cdot i e^{i\theta} d\theta$$

$$\therefore i^2 = -1$$

$\therefore$ using $u \cdot v$

$$= - \int_0^{2\pi} \theta e^{i\theta} d\theta$$

$$= - \left[\theta \cdot \frac{e^{i\theta}}{i} - 1 \cdot \left(\frac{e^{i\theta}}{i^2} \right) \right]_0^{2\pi}$$

$$= - \left[\frac{\theta \cdot e^{i\theta}}{i} + e^{i\theta} \right]_0^{2\pi}$$

$$= - \left[\left(\frac{2\pi \cdot e^{i2\pi}}{i} \right) - (0 + e^0) \right]$$

$$= - \left[\frac{2\pi e^{i2\pi}}{i} + e^{i2\pi} - 1 \right]$$

$$e^{i2\pi} = 1$$

$$= - \left[\frac{2\pi}{i} + 1 - 1 \right]$$

$$= - \frac{2\pi}{i} = \frac{-2\pi \cdot i}{i \times i} = \frac{-2\pi i}{i^2}$$

$$= \frac{-2\pi i}{-1} = 2\pi i \quad \text{proved}$$

Cauchy's Integral Theorem (Formula)

$$\int_C \frac{f(z)}{z-z_0} dz = 2\pi i f(z_0) \quad \int_C \frac{dz}{z-z_0} = 2\pi i$$

$$\int_C \frac{f(z)}{(z-z_0)^n} dz = \frac{2\pi i f^{(n-1)}(z_0)}{(n-1)!}$$

TYPE

i) $\int_C f(z) dz = \int_C \frac{\phi(z)}{(z-z_1)(z-z_2)} dz$ $[z_1, z_2 \text{ are both inside } C]$

ii) $\int_C f(z) dz = \int_C \frac{\phi(z)}{(z-z_1)(z-z_2)} dz$ $\begin{bmatrix} z_1 \text{ inside} \\ z_2 \text{ outside} \end{bmatrix}$

iii) $\int_C f(z) dz = \int_C \frac{\phi(z)}{(z-z_1)(z-z_2)} dz$ $[z_1, z_2 \text{ are both outside } C]$

NOTE: $|z| = 1 \rightarrow$ radius (r) passing through origin

Ex Evaluate $\int_C \frac{z+2}{(z-3)(z-4)} dz$ where C is the circle $|z|=1$

$$\int_C \frac{z+2}{(z-3)(z-4)} dz$$

$$\begin{array}{ll} z-3=0 & z-4=0 \\ z=3 & z=4 \end{array}$$

$$|z| = r = 2 = \text{radius}$$

Both the points lie outside the circle

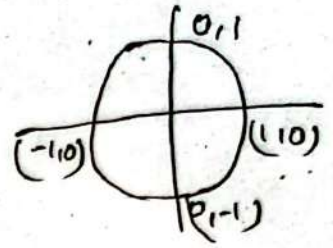
By C.T. theorem $\int_C \frac{z+1}{(z-3)(z-4)} dz = 0$

Evaluate $\int \frac{z+6}{z^2-4} dz$ where C is the circle.

I) $|z|=1$ II) $|z-2|=1$ III) $|z+2|=1$

$$\int \frac{z+6}{z^2-4} dz = \frac{z+6}{z^2-2^2}$$

$$= \int \frac{z+6}{(z+2)(z-2)} dz$$



$$z+2=0 \quad z-2=0$$

$$z=-2 \quad z=2$$

$|z|=1 = \text{radius}(r)$

Both the points lie in outside the circle
By C.T. theorem

$$\int \frac{z+6}{z^2-4} dz = 0$$

ii)

$$|z-2|=1$$

$$z=-2 \quad z=2$$

$$|z-2| = |-2-2| = |-4| = 4 \quad (\text{outside})$$

$$|z-2| = |2-2| = |0| = 0 \quad (\text{inside})$$

$$f(z) = \frac{z+6}{\text{outside}} = \frac{z+6}{z+2}$$

$$f(z) = \frac{z+6}{z+2} = \frac{8}{4} = 2$$

$$\int \frac{z+6}{z^2-4} dz = \int \frac{z+6}{(z+2)(z-2)} dz = 2\pi i f(z_0) = 2\pi i f(2)$$

$$= 2\pi i (2) = 4\pi i$$

$\therefore f(z_0)$ is taken from inside part

iii)

$$|z+2|=1$$

$$z = -2 \quad z = 2$$

$$\text{radius} = 1$$

$$|-2+2| = |0| = 0 \quad (\text{inside})$$

$$|2+2| = |4| = 4 \quad (\text{outside})$$

$$f(z) = \frac{z+6}{z-2}$$

$$f(-2) = \frac{-2+6}{-2-2} = \frac{4}{-4} = -1$$

$$\int \frac{z+6}{(z-2)(z+2)} dz = \oint 2\pi i f(z_0) = 2\pi i f(-2)$$

$$= 2\pi i (-1)$$

$$= -2\pi i$$

Evaluate $\int \frac{z+2}{z^3-2z^2} dz$ where C is the circle $|z-2-i|=2$

$$\int \frac{z+2}{z^2(z-2)} dz$$

$$z^2 = 0 \quad z-2=0$$

$$z = 0 \quad z = 2$$

$$\text{radius} = 2$$

$$z=0 \quad |z-2-i| = |(0-2)-i| = \sqrt{(-2)^2 + (-1)^2} = \sqrt{4+1} = \sqrt{5} = 2.2360 \quad [\text{outside}]$$

$$z=2 \quad |2-2-i| = |0-i| = \sqrt{0^2 + (-1)^2} = \sqrt{0+1} = \sqrt{1} = 1 \quad [\text{inside}]$$

$$f(z) = \frac{z+2}{z^2}$$

[Always taken from inside side part]

$$f(2) = \frac{2+2}{2^2} = \frac{4}{4} = 1$$

$$\int \frac{z+2}{z^3 - 2z^2} dz = \int \frac{z+2}{z^2(z-2)} dz$$

$$= 2\pi i f(2)$$

$$= 2\pi i (1)$$

$$= 2\pi i$$

Evaluate

$$\int_C \frac{dz}{z^3(z+4)} \text{ where } C \text{ is the circle } |z|=2$$

$$|z| = 2$$

$$\text{radius} = 2$$

$$z^3 = 0 \quad z+4 = 0$$

$$z = 0 \quad z = -4$$

↓
inside

↓
outside

$$f(z) = \frac{1}{z+4}$$

$$\frac{d}{dx} \frac{1}{x^n} = \frac{-n}{x^{n+1}}$$

$$f(z) = \frac{1}{z+4}$$

$$f'(z) = \frac{-1}{(z+4)^2}$$

$$f''(z) = \frac{(-1)(-2)}{(z+4)^3}$$

$$\therefore f''(0) = \frac{2}{0+4^3} = \frac{2}{64} = \frac{1}{32}$$

$$\int_C \frac{dz}{z^3(z+4)} = \int_C \frac{\frac{1}{z+4}}{z^3} dz$$

$$= \frac{2\pi i f^{(n-1)}(z_0)}{(n-1)!} = \frac{2\pi i f''(0)}{2}$$

$$= \pi i f''(0) = \pi i \times \frac{1}{32} = \frac{\pi i}{32}$$

Evaluate $\int_C \frac{\sin^6 z}{(z - \pi/6)^3} dz$ where C is $|z| = 1$.

Soln

$$z - \frac{\pi}{6} = 0$$

$$z = \pi/6 = 0.5235$$

$z = \pi/6$ lies inside

$$f(z) = \frac{\sin^6 z}{\text{outside}} = \sin^6 z$$

$$\int \frac{\sin^6 z}{(z - \pi/6)^3} dz = \frac{2\pi i f^{(3-1)}(\pi/6)}{(3-1)!}$$

$$= \frac{2\pi i f^2(\pi/6)}{2}$$

$$= \pi i f^2(\pi/6)$$

$$= \pi i \frac{21}{16}$$

$$= \frac{21\pi i}{16}$$

Ans

$$f(z) = \sin^6 z$$

$$f'(z) = 6[\sin^5 z \cdot \cos z]$$

$$f''(z) = 6[\sin^5 z \times (-\sin z) + (\cos z) 5 \cdot (\sin z)^4 \times \cos z]$$
$$= 6[-(\sin z)^5 + 5(\sin z)^4 (\cos z)]$$

$$f^2(\pi/6) = 6[-\sin(\pi/6)^5 + 5(\sin(\pi/6))^4 \cdot (\cos \pi/6)]$$

$$= 6\left[-\frac{1}{64} + 5\left(\frac{1}{16}\right)\left(\frac{3}{4}\right)\right]$$

$$= \frac{21}{16}$$

Q $\int_c \frac{e^{2z}}{(z-1)(z-2)} dz$ where c is the circle $|z|=3$

$$z-1=0 \quad z-2=0$$

$$z=1$$

$$z=2$$

↓
inside

↓
inside

[using partial fraction]

$$\int \frac{e^{2z}}{(z-1)(z-2)} dz = \frac{1}{(z-1)(z-2)} = \frac{A}{z-1} + \frac{B}{z-2}$$

$$1 = A(z-2) + B(z-1)$$

$$1 = A(z-2) \quad \text{put } z=2$$

$$1 = B(z-1)$$

$$1 = B(1)$$

$$1 = B$$

$$\text{put } z=1$$

$$1 = A(1-2)$$

$$1 = A(-1)$$

$$-1 = A$$

$$= \int e^{2z} \left(\frac{1}{(z-1)(z-2)} \right) dz$$

$$= \int e^{2z} \left(\frac{-1}{z-1} + \frac{1}{z-2} \right) dz$$

$$= \int e^{2z} \left(\frac{1}{z-2} - \frac{1}{z-1} \right) dz$$

$$= \int \frac{e^{2z}}{z-2} dz - \int \frac{e^{2z}}{z-1} dz$$

$$= 2\pi i f(2) - 2\pi i f(1)$$

$$= 2\pi i e^4 - 2\pi i e^2$$

$$= 2\pi i e^2 (e^2 - 1)$$

Ans

$$\begin{aligned} f(2) &= e^{2z} \\ &= e^{2(2)} \\ &= e^4 \end{aligned}$$

$$\begin{aligned} f(1) &= e^{2z} \\ &= e^2 \end{aligned}$$

x^2

$$\frac{x^2}{a} + \frac{y^2}{b} = 1$$

$$4x^2 + y^2 - 4y + 4 = 4$$

$$4x^2 + y^2 = 4y$$

Ex 3.3

3 (d) Integrate $\oint_C \frac{e^z}{z^2(z-1-i)} dz$, C consists $|z| \geq 2$ C.C.W and $|z| = 1$ C.W

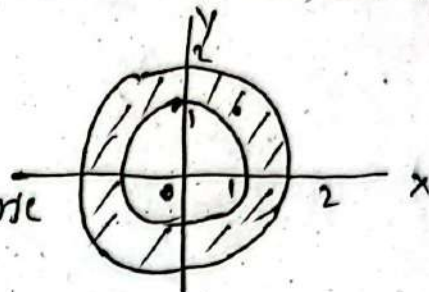
The given integrand is not analytic

For $z^2(z-1-i) \neq 0$

$z \neq 0 \quad z \neq 1+i \quad \{1, i\}$

↓
outside

↓
inside.



The point $z = 1+i$ or $(1,1)$ lies within the annulus
but $z = 0$ $(0,0)$ lies outside of the annulus
here, $z_0 = 1+i$

$$\oint_C \frac{e^z}{z^2(z-1-i)} dz = \oint_C \frac{\frac{e^z}{z^2}}{\{z-(1+i)\}} dz$$

$$= 2\pi i \cdot f(1+i) \text{ where } f(z) = \frac{e^z}{z^2}$$

$$= 2\pi i \times \frac{e^{(1+i)^2}}{(1+i)^2}$$

$$= 2\pi i \times \frac{e^{2i}}{2i} = \pi e^{2i}$$

(c) $\oint \frac{+4\pi z}{(z-\pi/4)^3} dz$ in $|z|=1$

f) $\oint \frac{z^3 \cdot e^z}{(z-1/2)^3} dz$ in $|z|=1$

2 (b) $\oint \frac{f(z) \log(z)}{\sqrt{(z-2)^2}} dz$ $C = |z-1|=2$

(c) $\oint_C \frac{\cosh 4z}{(z-4)^3} dz$ where
 C consists of $|z| \leq 6$ C.W
 $|z-3| = 2$ C.W

Taylor Series

$$f(z) = f(a) + (z-a) f'(a) + \frac{(z-a)^2}{2!} f''(a) + \frac{(z-a)^3}{3!} f'''(a) + \dots$$

Q Expand $f(z) = \sin z$ as a Taylor's series around $z = \pi/4$

$$\begin{aligned} f(z) &= \sin z \\ f'(z) &= \cos z \\ f''(z) &= -\sin z \\ f'''(z) &= -\cos z \end{aligned}$$

$$\begin{aligned} f(a) &= f\left(\frac{\pi}{4}\right) = \sin \pi/4 = \frac{1}{\sqrt{2}} \\ f'(a) &= f'\left(\frac{\pi}{4}\right) = \cos \pi/4 = \frac{1}{\sqrt{2}} \\ f''(a) &= f''\left(\frac{\pi}{4}\right) = -\sin \pi/4 = -\frac{1}{\sqrt{2}} \\ f'''(a) &= f'''\left(\frac{\pi}{4}\right) = -\cos \pi/4 = -\frac{1}{\sqrt{2}} \end{aligned}$$

According to Taylor's series,

$$\begin{aligned} f(z) &= f(a) + (z-a) f'(a) + \frac{(z-a)^2}{2!} f''(a) + \frac{(z-a)^3}{3!} f'''(a) + \dots \\ \sin z &= \frac{1}{\sqrt{2}} + (z - \pi/4) \cdot \frac{1}{\sqrt{2}} + \frac{(z - \pi/4)^2}{2!} \left(-\frac{1}{\sqrt{2}}\right) + \frac{(z - \pi/4)^3}{3!} \left(-\frac{1}{\sqrt{2}}\right) + \dots \\ \sin z &= \frac{1}{\sqrt{2}} \left[1 + (z - \pi/4) - \frac{(z - \pi/4)^2}{2!} - \frac{(z - \pi/4)^3}{3!} + \dots \right] \end{aligned}$$

H Show that for every finite value of z ,

$$e^z = e + e \leq \frac{(z-1)^n}{n!}$$

$$\begin{aligned} f(z) &= e^z \\ f'(z) &= e^z \\ f''(z) &= e^z \\ f'''(z) &= e^z \end{aligned}$$

$$\begin{aligned} f(a) &= f(1) = e^1 = e \\ f'(a) &= f'(1) = e^1 = e \\ f''(a) &= f''(1) = e^1 = e \\ f'''(a) &= f'''(1) = e^1 = e \end{aligned}$$

According to Taylor series.

$$f(z) = f(a) + (z-a)f'(a) + \frac{(z-a)^2}{2!} f''(a) + \frac{(z-a)^3}{3!} f'''(a) + \dots$$

$$e^z = e + (z-1)e + \frac{(z-1)^2}{2!} e + \frac{(z-1)^3}{3!} e + \dots$$

$$e^z = e + e \left[(z-1) + \frac{(z-1)^2}{2!} + \frac{(z-1)^3}{3!} + \dots \right]$$

$$e^z = e + e \sum \frac{(z-1)^n}{n!} \quad \underline{\text{proved}}$$

sequence and series

Some expansions

$$(1) \quad \frac{1}{1-z} = \sum_{n=0}^{\infty} z^n = 1 + z + z^2 + \dots \infty$$

$$(2) \quad \frac{1}{1+z} = \sum_{n=0}^{\infty} (-1)^n (z)^n = 1 - z + z^2 - z^3 + \dots \infty$$

$$(3) \quad e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} = 1 + \frac{z}{1!} + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots \infty$$

$$(4) \quad \sinh z = \sum_{n=0}^{\infty} \frac{(-1)^n \cdot z^{2n+1}}{(2n+1)!} = \frac{z}{1!} - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots \infty$$

$$(5) \quad \cosh z = \sum_{n=0}^{\infty} \frac{(-1)^n \cdot z^{2n}}{(2n)!} = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \dots \infty$$

$$(6) \quad \cosh z = \sum_{n=0}^{\infty} \frac{z^{2n}}{(2n)!} = 1 + \frac{z^2}{2!} + \frac{z^4}{4!} + \dots \infty$$

$$(7) \quad \sinh z = \sum_{n=0}^{\infty} \frac{z^{2n+1}}{(2n+1)!} = \frac{z}{1!} + \frac{z^3}{3!} + \frac{z^5}{5!} + \dots \infty$$

$$(8) \quad \log_e(1+z) = \frac{z}{1} - \frac{z^2}{2} + \frac{z^3}{3} - \frac{z^4}{4} + \dots = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} z^n}{n}$$

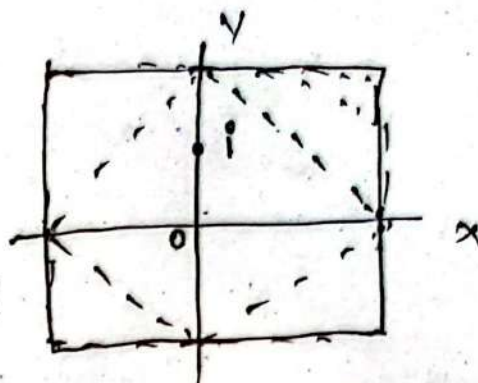
$$(9) \quad \log_e(1-z) = -z - \frac{z^2}{2} - \frac{z^3}{3} - \frac{z^4}{4} - \dots = - \sum_{n=1}^{\infty} \frac{z^n}{n}$$

* Evaluate $\oint_C \left[\frac{z^3 + \sin z}{(z-i)^3} \right] dz$ where C is the boundary of the square with vertices $\pm 2, \pm 2i$
 so in

$$\oint_C \frac{z^3 + \sin z}{(z-i)^3} dz$$

This integrand is not analytic.

For $z-i=0$ i.e. $z=i = (0,1)$



clearly i lies within the given square (region)
 we use derivative of Cauchy's integral formula
 $z = z_0 = i$

$$\oint_C \frac{(z^3 + \sin z)}{(z-i)^3} dz = \frac{2\pi i}{2!} f''(i) \quad \text{where } f(z) = z^3 + \sin z$$

$$f'(z) = 3z^2 + \cos z$$

$$f''(z) = 6z - \sin z$$

$$f''(i) = 6i - \sin i$$

$$= \frac{2\pi i}{2!} \times 6i - \sin i$$

$$= \pi i (6i - \sin i)$$

$\sum$

* Evaluate $\oint_C \frac{z^6}{(2z-1)^6} dz$ where C is a unit circle

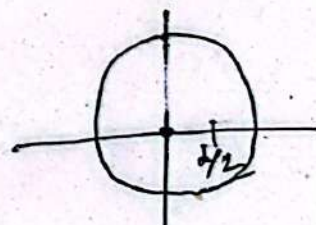
so in

$$\frac{1}{2^6} \oint_C \frac{z^6}{(z-\frac{1}{2})^6} dz$$

This integrand is not analytic

For $2z-1=0$

$$z = \frac{1}{2}$$



$$\frac{1}{2^6} \oint_C \frac{z^6}{(2z - \frac{1}{2})^6} dz = \frac{2\pi i}{2^6} \times \frac{2\pi i}{5!} f^{(5)}(\frac{1}{2})$$

where $f(z) = z^6$

$$f'(z) = 6z^5$$

$$f''(z) = 30z^4$$

$$f'''(z) = 120z^3$$

$$f^{(4)}(z) = 360z^2$$

$$f^{(5)}(z) = 720z$$

$$f^{(5)}(\frac{1}{2}) = 720 \times \frac{1}{2} = 360$$

$$= \frac{2\pi i}{5 \times 4 \times 3 \times 2} \times 360 \times \frac{1}{2^6}$$

$$= \frac{3\pi i}{32}$$

Q $\oint_C \frac{z+4}{z^2+2z+5} dz$ where $C: |z+1-i| = 2$

soln

$$z^2 + 2z + 5 = 0$$

$$\text{or } z^2 + 2z + 1 + 4 = 0$$

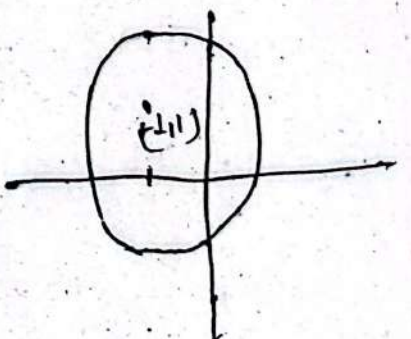
$$(z+1)^2 - (2i)^2 = 0$$

$$(z+1+2i)(z+1-2i) = 0$$

$$z = -1-2i, \text{ and } z = -1+2i$$

↓
outside

↓
inside



Q

$$\frac{z+4}{(z+1+2i)(z+1-2i)} = \frac{A}{z+1+2i} + \frac{B}{z+1-2i}$$

$$z+4 = A(z+1-2i) + B(z+1+2i)$$

$$\text{Put } z = -1+2i$$

$$\text{or, } -1+2i+4 = B(-1+2i+1+2i)$$

$$\text{or, } 3+2i = 4iB$$

$$B = \frac{3+2i}{4i} = \frac{2i}{4i} + \frac{3}{4i} = \frac{(2-3i)}{4}$$

$$\text{Similarly, } A = \frac{(2+3i)}{4}$$

$$\frac{2-i}{4} =$$

$$\int_C \frac{\frac{2+3i}{4}}{z+1+2i} dz + \int_C \frac{\frac{2-3i}{4}}{z+1-2i} dz$$

$$0 + \int_C \frac{2-3i}{4(z+1-2i)} dz$$

$$2\pi i + \frac{2-3i}{4} \int_C \frac{1}{(z+1-2i)} dz$$

$$= \frac{2-3i}{4} 2\pi i \quad [\text{by C.T theorem}]$$

$$= \frac{\pi i (4-6i)}{2}$$

h

Sequence and series

2.) Find Maclaurin's series of

(P) $f(z) = \cos 2z^2$

we have $\cos z = \sum_{n=0}^{\infty} \frac{(-1)^n \cdot z^{2n}}{(2n)!}$

$$\begin{aligned} f(z) = \cos 2z^2 &= \sum_{n=0}^{\infty} \frac{(-1)^n (2z^2)^{2n}}{(2n)!} \\ &= 1 - \frac{(2z^2)^2}{2!} + \frac{(2z^2)^4}{4!} - \dots \\ &= 1 - 2z^4 + \frac{2z^8}{3} - \dots \end{aligned}$$

(II) $f(z) = \sin^2 z$

$$= \frac{1 - \cos 2z}{2}$$

$$= \frac{1}{2} - \frac{1}{2} \sum_{n=0}^{\infty} \frac{(-1)^n (2z)^{2n}}{(2n)!}$$

$$= \frac{1}{2} - \frac{1}{2} \left[1 - \frac{4z^2}{2!} + \frac{16z^4}{4!} - \dots \right]$$

(C) $f(z) = \frac{z+2}{1-z^2} = \frac{z+2}{(1+z)(1-z)}$

here, $\frac{z+2}{(1+z)(1-z)} = \frac{A}{1+z} + \frac{B}{1-z}$

$$z+2 = A(1-z) + B(1+z) \quad \text{--- (1)}$$

put $z = -1$ in (1) $\Rightarrow B = 3/2$

similarly $A = 1/2$

$$f(z) = \frac{1/2}{1+z} + \frac{3/2}{1-z}$$

$$= \frac{1}{2} \cdot \frac{1}{1+z} + \frac{3}{2} \cdot \frac{1}{1-z}$$

$$= \frac{1}{2} \sum_{n=0}^{\infty} (-1)^n z^n + \frac{3}{2} \sum_{n=0}^{\infty} z^n$$

$$= \frac{1}{2} [1 - z + z^2 - z^3 + \dots] + \frac{3}{2} (1 + z + z^2 + z^3 + \dots)$$

$$= 2 + z + 2z^2 + z^3 + \dots$$

Σ

9) $f(z) = \frac{1}{z^2+1} = \frac{1}{z^2-1}$

$$= \frac{1}{(z+i)(z-i)} = \frac{A}{z+i} + \frac{B}{z-i}$$

on solving, $A = -\frac{1}{2i}$ and $B = \frac{1}{2i}$

The ① becomes.

$$f(z) = -\frac{1}{2i} \cdot \frac{1}{z+i} + \frac{1}{2i} \cdot \frac{1}{z-i}$$

$$= -\frac{1}{2i} \left[-\frac{1}{z+i} + \frac{1}{z-i} \right]$$

$$= \frac{1}{2i} \cdot \frac{1}{i} \left[-\frac{1}{1+z/i} - \frac{1}{1-z/i} \right]$$

$$= \frac{1}{2} \left[\frac{1}{1+z/i} + \frac{1}{1-z/i} \right]$$

$$= \frac{1}{2} \left[\sum_{n=0}^{\infty} (-1)^n \left(\frac{z}{i}\right)^n + \left(\frac{z}{i}\right)^n \right]$$

$$= \frac{1}{2} \left(\frac{z}{i}\right)^n [(-1)^n + 1]$$

Σ

$$= \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{z}{i}\right)^n [(-1)^n + 1]$$

$$= \frac{1}{2} [2 - 2z^2 + 2z^4 + \dots]$$

2. Find Taylor's expansion of

1) $f(z) = \frac{1}{z}$ at $z = 2$

we have Taylor's series of $f(z)$ at $z = a$ is

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n \quad \text{where } a_n = \frac{f^{(n)}(a)}{n!}$$

here $a = 2$ given,

here, $f(z) = \frac{1}{z}$

$$f'(z) = \frac{1}{z^2} = \frac{(-1)^1}{z^{1+1}} = \frac{(-1)^1 \cdot 1!}{z^{1+1}}$$

$$f''(z) = \frac{2}{z^3} = \frac{(-1)^2 \cdot 2 \times 1}{z^{2+1}} = \frac{(-1)^2 \cdot 2!}{z^{2+1}}$$

$$f'''(z) = \frac{-6}{z^4} = \frac{(-1)^3 \cdot 3 \times 2 \times 1}{z^{3+1}} = \frac{(-1)^3 \cdot 3!}{z^{3+1}}$$

$$f^{(n)}(z) = \dots = \frac{(-1)^n \cdot n!}{z^{n+1}}$$

$$f^{(n)}(z) = \frac{(-1)^n n!}{z^{n+1}}$$

$$\therefore f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n \cdot n!}{n! \cdot z^{n+1}} \cdot (z-2)^n$$

$$= \sum_{n=0}^{\infty} (-1)^n \frac{(z-2)^n}{z^{n+1}}$$

$$= -\frac{1}{2} - \frac{(z-2)}{4} + \frac{(z-2)^2}{8}$$

c) Expand $f(z) = \sinh(2z-i)$ at $z = \frac{i}{2}$

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n \quad \text{where } a = \frac{i}{2}$$

$$f(z) = \sinh(2z-i)$$

$$f(z) = 2 \cosh(2z-i) = 2 \sinh(2z-i + \frac{\pi}{2})$$

$$f'(z) = 2^2 \cosh(2z-i + \frac{\pi}{2}) = 2^2 \sinh(2z-i + \frac{2\pi}{2})$$

$$f^{(n)}(z) = \dots = 2^n \sinh(2z-i + \frac{n\pi}{2})$$

$$\therefore f^{(n)}(\frac{i}{2}) = 2^n \sinh(2 \cdot \frac{i}{2} - i + \frac{n\pi}{2}) = 2^n \cdot \sinh \frac{n\pi}{2}$$

$$a_n = \frac{f^{(n)}(a)}{n!} = \frac{2^n \sinh(\frac{n\pi}{2})}{n!}$$

The required Taylor's series is

$$f(z) = \sum_{n=0}^{\infty} \frac{2^n \sinh(\frac{n\pi}{2})}{n!} \cdot (z - \frac{i}{2})^n$$

$$= 2(z - \frac{i}{2}) + \frac{8}{3!}(z - \frac{i}{2})^3 + \frac{32}{5!}(z - \frac{i}{2})^5 + \dots$$

3) write Taylor's expansion for

b) $f(z) = \frac{1}{z^2 + 2z + 3}$ for $|z| < 1$

we have $\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n$

here $f(z) = \frac{1}{z^2 + 2z + 1 + 2}$

$$= \frac{1}{(z+1)^2 - (\sqrt{2}i)^2} = \frac{1}{(z+1+\sqrt{2}i)(z+1-\sqrt{2}i)}$$

$$\frac{A}{z+1+\sqrt{2}i} + \frac{B}{z+1-\sqrt{2}i}$$

$$\text{or, } 1 = A(z+1-\sqrt{2}i) + B(z+1+\sqrt{2}i) \quad \dots (1)$$

$$\text{Solving for } A = \frac{1}{2\sqrt{2}i}$$

$$B = \frac{1}{2\sqrt{2}i}$$

$$f(z) = \frac{1}{2\sqrt{2}i} \left[\frac{1}{z+1-\sqrt{2}i} - \frac{1}{z+1+\sqrt{2}i} \right]$$

$$= \frac{1}{2\sqrt{2}i} \left[\frac{1}{(1-\sqrt{2}i)} \left\{ 1 + \frac{z}{1-\sqrt{2}i} \right\} - \frac{1}{(1+\sqrt{2}i)} \left\{ 1 + \frac{z}{1+\sqrt{2}i} \right\} \right]$$

$$= \frac{1}{2\sqrt{2}i} \left[\frac{1}{(1-\sqrt{2}i)} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z}{1-\sqrt{2}i} \right)^n - \frac{1}{1+\sqrt{2}i} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z}{1+\sqrt{2}i} \right)^n \right]$$

$$= \frac{1}{2\sqrt{2}i} \sum_{n=0}^{\infty} \left[\frac{(-1)^n \cdot z^n}{(1-\sqrt{2}i)^{n+1}} - \frac{(-1)^n \cdot z^n}{(1+\sqrt{2}i)^{n+1}} \right]$$

$$f(z) = \frac{1}{2\sqrt{2}i} \sum_{n=0}^{\infty} (-1)^n z^n \left[\frac{1}{(1-\sqrt{2}i)^{n+1}} - \frac{1}{(1+\sqrt{2}i)^{n+1}} \right]$$

$$c) \quad f(z) = \frac{z^3 + 2z^2}{z^2 + 2z + 3} \quad \text{if } |z| < \frac{1}{2}$$

$$f(z) = z - \frac{3z}{(z+1+\sqrt{2}i)(z+1-\sqrt{2}i)}$$

4(c) obtain the Taylor's expansion of

$$f(z) = \frac{z^3 + 1}{z^2 + z} \text{ about } z=1 \text{ and}$$

obtain the region of validity also

$$f(z) = \frac{z^3 + 1}{z^2 + z}$$

Here, $z-1=0$
Then, we need of $(z-1)$ in series

$$\begin{array}{r} \text{then, } z^2 + z \overline{) 2z^3 + 1} \quad (2z-2) \\ \underline{-2z^3 \quad + 2z^2} \\ -2z^2 + 1 \\ \underline{+ 2z^2 \quad + 2z} \\ 2z + 1 \end{array}$$

$$\therefore f(z) = (2z-2) + \frac{(2z+1)}{z(z+1)}$$

$$= 2(z-1) + 2(-1) + \frac{A}{z} + \frac{B}{z+1}$$

$$\frac{2z+1}{z(z+1)} = \frac{A(z+1) + Bz}{z(z+1)}$$

put $z=0$, $A=1$, put $z=-1$, $B=2$

$$f(z) = 2(z-1) + 2(-1) + \frac{1}{z} + \frac{2}{z+1}$$

$$= 2(z-1) + 2(-1) + \frac{1}{(z-1)+1} + \frac{2}{z-1+1+1}$$

$$= 2(z-1) + 2(-1) + \frac{1}{1} \left[\frac{1}{1 + \frac{z-1}{1}} \right] + \frac{1}{1+i} \left[\frac{1}{1 + \frac{z-1}{1+i}} \right]$$

$$= 2(z-1) + 2(-1) + \frac{1}{1} \sum_{n=0}^{\infty} (-1)^n \cdot \frac{(z-1)^n}{1^n} + \frac{1}{1+i} \sum_{n=0}^{\infty} \frac{(z-1)^n}{(1+i)^n}$$

$$= 2(z-1) + 2(-1) + \sum_{n=0}^{\infty} \left[\frac{1}{1^{n+1}} + \frac{2}{(1+i)^{n+1}} \right] (-1)^n (z-1)^n$$

This is the required expansion of $f(z)$ about $z=1$,

For validity of expansion.

$$\left| \frac{z-q}{p} \right| < 1$$

$$\text{or, } |z-q| < 1$$

$\therefore$ valid region

$$\left| \frac{z-q}{1+q} \right| < 1$$

$$\Rightarrow |z-q| < \sqrt{2}$$

$\therefore$ Not the valid region

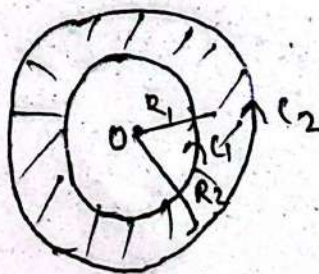
State Laurent series

If $f(z)$ be analytic on two concentric circles C_1 & C_2 with centre at a also analytic in annular region A , then $f(z)$ can be expressed as a convergent series of positive and negative power of $(z-a)$ in the form $f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=1}^{\infty} b_n (z-a)^{-n}$

$$\text{where } a_n = \frac{1}{2\pi i} \oint_C \frac{f(w)}{(w-a)^{n+1}} dw, \quad n=0, 1, 2, 3, \dots$$

$$\& b_n = \frac{1}{2\pi i} \oint_C \frac{f(w)}{(w-a)^{-n+1}} dw, \quad n=1, 2, 3, \dots$$

and C being simple closed path lying within the annular region enclosing the inner boundary of annular region A .



$$\therefore r_1 < |z-a| < r_2$$

Ex 5.2

2. obtain the Laurent series in $0 < |z| < R$

(a) $\frac{\cos z}{z^4}$

$$f(z) = \frac{\cos z}{z^4}$$

∴ Laurent series of $f(z)$

$$= \frac{1}{z^4} \sum_{n=0}^{\infty} \frac{(-1)^n \cdot z^{2n}}{(2n)!}$$

$$= z^{-4} - \frac{z^{-2}}{3} + \frac{1}{24} - \frac{z^2}{720} + \dots \infty$$

d) $f(z) = \frac{e^{-\frac{1}{2}z^2}}{z^2}$

we know, $e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$

$$\therefore f(z) = \frac{1}{z^2} \sum_{n=0}^{\infty} \frac{\left(-\frac{1}{2}z^2\right)^n}{n!}$$

$$= z^{-2} - z^{-4} + \frac{z^{-6}}{2} - \frac{z^{-8}}{6} + \dots$$

2) Find the Laurent series that converges for $0 < |z - z_0| < R$ of

9) $f(z) = \frac{1}{z^2+1}$ at $z_0 = i = z$

$$f(z) = \frac{1}{z^2+1} = \frac{1}{(z+i)(z-i)}$$

we observe the series for $|z-i| < 1$
i.e. $0 < |z-i| < 1$

$$f(z) = \frac{1}{(z-i)} \cdot \frac{1}{z-i+2i}$$

$$= \frac{1}{z-i} \cdot \frac{1}{2i \left(1 + \frac{z-i}{2i}\right)} = \frac{1}{z-i} \sum_{n=0}^{\infty} \frac{(-1)^n (z-i)^n}{(2i)^{n+1}}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n (z-i)^{n-1}}{(2i)^{n+1}}$$

$$= -\frac{1}{2} (z-i)^{-1} + \frac{1}{4} + \frac{i}{8} (z-i) - \frac{(z-i)^2}{16} + \dots$$

d) $f(z) = \frac{\cosh z}{(z+\pi i)^2}$

For $|z+\pi i| < 1$

here, $z_0 = -\pi i$ we observe the series for power of $(z+\pi i)$

$$\therefore f(z) = \frac{e^z + e^{-z}}{2(z+\pi i)^2}$$

$$= \frac{e^{z+\pi i - \pi i} + e^{-(z+\pi i) + \pi i}}{2(z+\pi i)^2}$$

$$\begin{aligned} e^{\pi i} &= \cos \pi + i \sin \pi = -1 \\ e^{-\pi i} &= \cos(-\pi) + i \sin(-\pi) = -1 \end{aligned}$$

$$= \frac{e^{(z+\pi i)} \cdot (-1) + e^{-(z+\pi i)} \cdot (-1)}{2(z+\pi i)^2}$$

$$= \frac{(-1)}{(z+\pi i)^2} \left[\frac{e^{z+\pi i} + e^{z-\pi i}}{2} \right]$$

$$= \frac{-2}{(z+\pi i)^2} \cosh(z + \pi i)$$

2) obtain the expansion of $f(z) = \frac{z-1}{z^2}$ in the power of $(z-1)$ Indicate the region of validity.
 Here, power of $z-1$ means the expansion is about $z=1$

Then, $|z-1| < 1$

Then, ~~power~~ series of $\frac{z}{z^2}$ is $\sum_{n=0}^{\infty} a_n (z-1)^n$ where

$$a_n = \frac{f^{(n)}(1)}{n!}$$

$$\text{Then } f(z) = \frac{z}{z^2}$$

$$f'(z) = \frac{-2}{z^3} = \frac{(-1)^2 (1+1)!}{z^{2+1}}$$

$$f''(z) = \frac{6}{z^4} = \frac{(-1)^3 (2+1)!}{z^{2+2}}$$

$$\vdots$$

$$f^{(n)}(z) = \dots = \frac{(-1)^n (n+1)!}{z^{n+2}}$$

$$f^{(n)}(1) = \frac{(-1)^n (n+1)!}{1}$$

$$\therefore a_n = \frac{f^{(n)}(1)}{n!} = \frac{(-1)^n (n+1)!}{n!} = (-1)^n (n+1)$$

$$\text{Then, } \frac{z}{z^2} = \sum_{n=0}^{\infty} (-1)^n (n+1) (z-1)^n$$

which is ~~valid~~ valid for $|z-1| < 1$

$$\text{If } |z-1| > 1$$

Then Laurent series of $\frac{1}{z^2}$ is

$$\frac{z-1}{z^2} = z-1 \sum_{n=0}^{\infty} (-1)^n (n+1) (z-1)^{-n+1}$$

$$\frac{z-1}{z^2} = (z-1) \sum_{n=0}^{\infty} (-1)^n (n+1) \frac{1}{(z-1)^n}$$

$$= \sum_{n=0}^{\infty} (-1)^n (n+1) (z-1)^{-n+1}$$

valid for $|z-1| > 1$

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Find Laurent expansion of

$$f(z) = \frac{z^2-1}{z^2+5z+6} \quad \text{in the region } 2 < |z| < 3$$

$$\begin{array}{r} z^2+5z+6 \overline{) z^2-1} \\ \underline{z^2+5z+6} \\ -5z-7 \end{array}$$

$$f(z) = 1 - \frac{(5z+7)}{(z+2)(z+3)}$$

$$\text{here, } \frac{5z+7}{(z+2)(z+3)} = \frac{A}{(z+2)} + \frac{B}{(z+3)}$$

$$5z+7 = A(z+3) + B(z+2) \quad \dots (1)$$

on solving $A = -3, B = 8$

$$f(z) = 1 + \frac{3}{z+2} - \frac{8}{z+3}$$

we have $2 < |z| < 3$

$$\therefore f(z) = 1 + 3 \left[\frac{1}{2(1+z/2)} \right] - 8 \left[\frac{1}{3(1+z/3)} \right]$$

$$= 1 + \frac{3}{2} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z}{2} \right)^n - \frac{8}{3} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z}{3} \right)^n$$

$$= 1 + 3 \sum_{n=0}^{\infty} (-1)^n \frac{z^n}{2^{n+1}} - 8 \sum_{n=0}^{\infty} (-1)^n \frac{z^n}{3^{n+1}}$$

Laurent series

Imp Expand $f(z) = \frac{7z-2}{z^3-z^2-2z}$ in the region

- a) $0 < |z+1| < 1$ b) $1 < |z+1| < 3$ c) $|z+1| > 3$

here, $f(z) = \frac{7z-2}{z(z+1)(z-2)} = \frac{A}{z} + \frac{B}{z+1} + \frac{C}{z-2}$

$$7z-2 = A(z+1)(z-2) + Bz(z-2) + Cz(z+1) \quad \text{--- (1)}$$

solving $A=2$, $B=-3$ and $C=2$ [using partial fractions]

$$f(z) = \frac{2}{z} - \frac{3}{z+1} + \frac{2}{z-2}$$

we have to arrange the terms suitably to get power of $(z+1)$ and $|z+1| < 1$.

Then $\therefore f(z) = \frac{1}{z+1-1} - \frac{3}{z+1} + \frac{2}{z+1-3}$

$$= -\frac{3}{z+1} - \frac{1}{1-(z+1)} - \frac{2}{3(1-\frac{z+1}{3})}$$

$$= \frac{-3}{z+1} - \sum_{n=0}^{\infty} (z+1)^n - \frac{2}{3} \sum_{n=0}^{\infty} \left(\frac{z+1}{3}\right)^n$$

$$= \frac{-3}{z+1} - \sum_{n=0}^{\infty} (z+1)^n - 2 \sum_{n=0}^{\infty} \frac{(z+1)^n}{3^{n+1}} \quad \text{is the}$$

required expansion.

(b) In the region $1 < |z+1| < 3$

$$f(z) = \frac{1}{z} - \frac{3}{z+1} + \frac{2}{z-2}$$

$$f(z) = \frac{-3}{z+1} + \frac{1}{z+1-1} + \frac{2}{z+1-3}$$

$$= \frac{-3}{z+1} - \frac{1}{(z+1)} \left[\frac{1}{1 - \frac{1}{z+1}} \right] - \frac{2}{3} \left[\frac{1}{1 - \frac{z+1}{3}} \right]$$

$$= \frac{-3}{z+1} - \frac{1}{z+1} \sum_{n=0}^{\infty} \left(\frac{1}{z+1}\right)^n - \frac{2}{3} \sum_{n=0}^{\infty} \left(\frac{z+1}{3}\right)^n$$

$$\therefore \left| \frac{1}{z+1} \right| < 1 \Rightarrow 1 < \frac{|z+1|}{3} < 1 \Rightarrow |z+1| < 3$$

(c) In the region $|z+1| > 3$

$$f(z) = \frac{1}{z} - \frac{3}{z+1} + \frac{2}{z-2}$$

$$= \frac{-3}{z+1} + \frac{1}{z} + \frac{2}{z-2}$$

$$= \frac{-3}{z+1} + \frac{1}{(z+1)-1} + \frac{2}{(z+1)-3}$$

$$= \frac{-3}{z+1} + \left(\frac{1}{z+1}\right) \left[\frac{1}{1 - \frac{1}{z+1}} \right] + \frac{2}{z+1} \left[\frac{1}{1 - \frac{3}{z+1}} \right]$$

$$= \frac{-3}{z+1} + \frac{1}{z+1} \sum_{n=0}^{\infty} \left(\frac{1}{z+1}\right)^n + \frac{2}{z+1} \sum_{n=0}^{\infty} \left(\frac{3}{z+1}\right)^n$$

$$= \frac{-3}{z+1} + \sum_{n=0}^{\infty} \frac{1}{(z+1)^{n+1}} + 2 \sum_{n=0}^{\infty} \frac{3^n}{(z+1)^{n+1}} \quad \underline{\underline{\quad}}$$

validity $\left| \frac{1}{z+1} \right| < 1 \Rightarrow 1 < |z+1| \Rightarrow \text{True.}$

and $\left| \frac{3}{z+1} \right| < 1 \Rightarrow 3 < |z+1| \Rightarrow \text{True}$

Conformal mapping

Bilinear Transformation (BLT)

Any transformation of the type $f(z) = \frac{az+b}{cz+d}$ where a, b, c, d are real or complex constants is said to be a bilinear transformation / Möbious transformation. [Note $ad-bc \neq 0$]

* Cross ratio property of BLT

When the points of z -plane say z_1, z_2, z_3 are given and also the corresponding points of w -plane w_1, w_2, w_3 are also given, then the associated BLT is given

$$\text{by, } \frac{(z-z_1)(z_2-z_3)}{(z_2-z_1)(z_3-z)} = \frac{(w-w_1)(w_2-w_3)}{(w_2-w_1)(w_3-w)}$$

$$f(z) = \frac{az+b}{cz+d}, \quad ad-bc \neq 0$$

$$\text{If } ad-bc = 0$$

$$\text{or, } ad=bc$$

$$\therefore \frac{a}{c} = \frac{b}{d} = k$$

$$a = ck$$

$$b = dk$$

$$\text{Then } f(z) = \frac{ckz+dk}{cz+d} = k \quad \text{constant function.}$$

"Not a transformation" That's where $ad-bc \neq 0$

Q Find the bilinear transformation which maps $z = 0, 1, \infty$ on to $w = -5, -1, 3$

$$\text{here } \begin{array}{l|l} z_1 = 0 & w_1 = -5 \\ z_2 = 1 & w_2 = -1 \\ z_3 = \infty & w_3 = 3 \end{array}$$

We have the cross-ratio format.

$$\frac{(z - z_1)(z_2 - z_3)}{(z_1 - z_2)(z_3 - z)} = \frac{(w - w_1)(w_2 - w_3)}{(w_1 - w_2)(w_3 - w)} \quad \text{Divide by } z_3$$

$$\text{or, } \frac{(z - z_1) \left(\frac{z_2}{z_3} - 1 \right)}{(z_1 - z_2) \left(1 - \frac{z}{z_3} \right)} = \frac{(w - w_1)(w_2 - w_3)}{(w_1 - w_2)(w_3 - w)}$$

$$\text{or, } \frac{(z - 0)(0 - 1)}{(0 - 1)(1 - 0)} = \frac{(w + 5)(-1 - 3)}{(-5 + 1)(3 - w)}$$

$$\text{or, } z = \frac{w + 5}{3 - w} \quad (z \text{ interval of } w)$$

$$\text{or, } 3z - wz = w + 5$$

$$\text{or, } 3z - 5 = w(1 + z)$$

$$w = \frac{3z - 5}{1 + z} = f(z) \text{ in the form of } \frac{az + b}{cz + d}$$

$$\text{here, } ad - bc \neq 0$$

Pracise

- (1) Find the BLT that maps $z = i, -i, 2$ into $w = 0, 1, \infty$
- (2) Find the BLT that maps $z = 1, -1, \infty$ into $w = 2+i, 1-i, 2$

2nd type

show that the bilinear transformation $w = \frac{2z+3}{z-4}$ transforms the circle $x^2+y^2=4x$ in the z -plane to the line $4y+3=0$ in w plane [also sketch the figure].

here, $x^2-4x+y^2=0$

$x^2-4x+4+(y-0)^2=4$

or, $(x-2)^2+(y-0)^2=2^2$

centre $(2,0)$ and radius $(r)=2$

$\Rightarrow |z-2|=2$

given $w = \frac{2z+3}{z-4}$

or, $zw - 4w = 2z + 3$

or, $(z-2)w =$

$z = \frac{4w+3}{w-2}$

or, $|z-2| = \left| \frac{4w+3}{w-2} - 2 \right|$

or, $2 = \left| \frac{2w+7}{w-2} \right|$

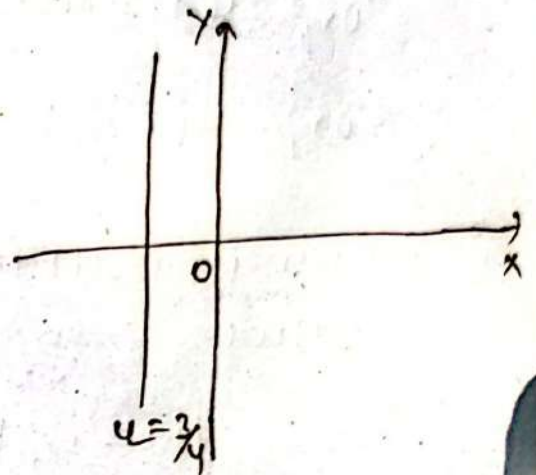
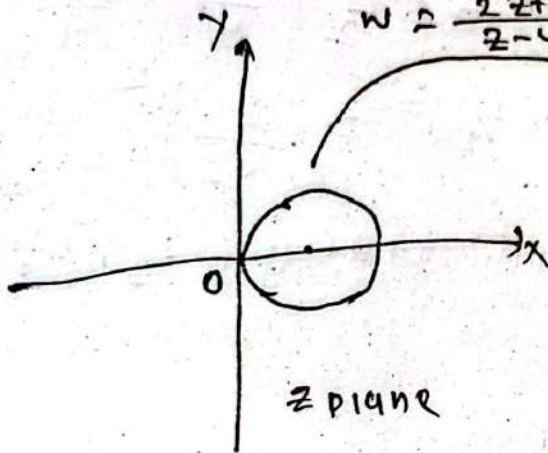
$2 = \frac{|2(4+iv)+7|}{|4+iv-2|}$

$$\text{or, } 2 = \frac{\sqrt{(2u+7)^2 + (2v)^2}}{\sqrt{(u-2)^2 + v^2}}$$

$$\text{or, } 4 = \frac{4u^2 + 28u + 49 + 4v^2}{u^2 - 4u + 4 + v^2}$$

on simplifying we get $4u + 3 = 0$ line or

$$u = \frac{2z+7}{z-4}$$



g) Find the image of the line $y - x + 1 = 0$ in z -plane under the mobius transformation.

$w = \frac{1}{z}$ also sketch figure

here, $w = \frac{1}{z}$

or, $z = \frac{1}{w}$

$$x + iy = \frac{1}{u + iv} \times \frac{u - iv}{u - iv} = \frac{u - iv}{u^2 + v^2}$$

$$x + iy = \frac{u}{u^2 + v^2} + i \left(\frac{-v}{u^2 + v^2} \right)$$

$$x = \frac{u}{u^2 + v^2} \quad \text{and} \quad y = \frac{-v}{u^2 + v^2}$$

substitute the value of x and y in the,

$$y - x + 1 = 0$$

$$\text{or, } \frac{-u}{u^2+u^2} - \frac{u}{u^2+u^2} + 1 = 0$$

$$\text{or, } u^2 + u^2 - u - u = 0$$

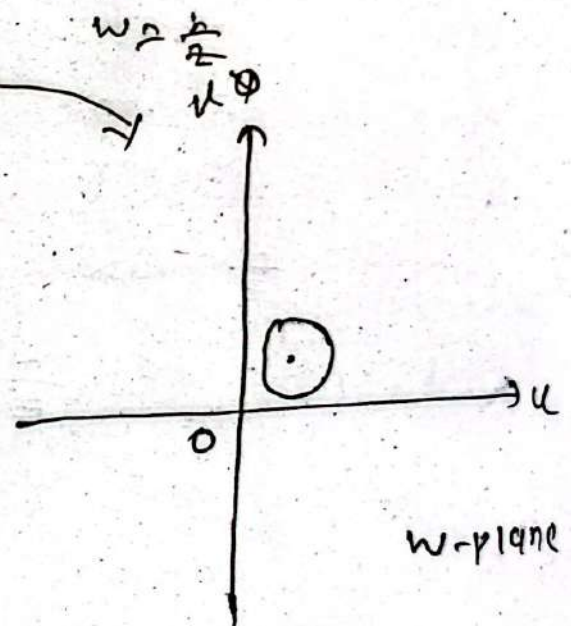
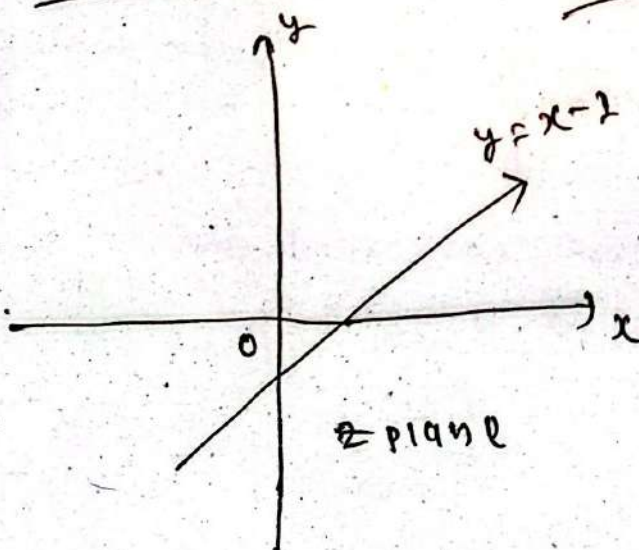
$$\text{or, } u^2 - 2 \cdot u \cdot \frac{1}{2} + u^2 - 2 \cdot u \cdot \frac{1}{2} + \frac{1}{4} = \frac{1}{4} + \frac{1}{4}$$

$$\text{or, } (u - \frac{1}{2})^2 + (u - \frac{1}{2})^2 = \left(\frac{1}{\sqrt{2}}\right)^2$$

which is a circle with centre $(\frac{1}{2}, \frac{1}{2})$ and
radius $= \frac{1}{\sqrt{2}}$

$$|w - \frac{1}{2}| = \frac{1}{\sqrt{2}}$$

Sketch



Chapter - 6

singularity, A function zeros and pole of a function

singularity A function $f(z)$ is said to be singular or has a singularity at point $z = z_0$ if $f(z)$ is not analytic at $z = z_0$ (perhaps not even defined at $z = z_0$) ; but $f(z)$ is analytic at every nbd of $z = z_0$

Example $f(z) = \frac{z}{2z-1}$ is not defined (or analytic) at $2z-1=0$ i.e. $z = \frac{1}{2} = z_0$

$\therefore f(z)$ has a singularity at $z = \frac{1}{2}$

Types of singularity

The Laurent series of $f(z)$ can be written as about $z = a$

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=1}^{\infty} b_n (z-a)^{-n} \dots \textcircled{1} \text{ is valid}$$

in the deleted nbd of $z = a$ except at $z = a$ itself (i.e. in the region of the form $0 < |z-a| < R$).

The 2nd part $\sum_{n=1}^{\infty} \frac{b_n}{(z-a)^n}$ is called principal part of $f(z)$ at isolated singularity at $z = a$:

There are 3 possibilities here

- (i) All coeff of b_n may be zero.
- (ii) Finite no. of terms in principal part.
- (iii) Infinite no. of terms in principal part.

Isolated singularity

Let $f(z)$ be a complex function, then the point $z = z_0$ is said to be isolated singularity if z_0 has a nbd without further singularity of $f(z)$, otherwise non-isolated singularity

$f(z) = \tan z \Rightarrow f(z)$ has isolated singularity at $z = \pm \pi/2, \pm 3\pi/2, \dots$

b) Removable singularity:

When the coeff of 0^n are zero, then we have

$$f(z) = \sum_{n=2}^{\infty} a_n (z-a)^n$$

which is analytic for $|z-a| < R$ and still not analytic at $z=a$

By redefining f at a , the singularity disappears. Such type of singularity is removable singularity. $z=a$ is said to be removable singular point if $\lim_{z \rightarrow a} f(z)$ exists.

$$\text{Let } f(z) = \frac{\sin(z-a)}{(z-a)}$$

$$= \frac{1}{(z-a)} \left[(z-a) - \frac{(z-a)^3}{3!} + \frac{(z-a)^5}{5!} - \dots \right]$$

$$= 1 - \frac{(z-a)^2}{3!} + \frac{(z-a)^4}{5!} - \dots$$

has no -ve power of $(z-a)$

redefines $f(z)$

$$\lim_{z \rightarrow a} \frac{\sin(z-a)}{(z-a)} = 1$$

so, $z = a$ is removable singularity

c) essential singularity

If the principal part of $f(z) = \sum_{n=20}^{\infty} a_n (z-a)^n + \sum_{n=1}^{\infty} b_n (z-a)^{-n}$ has infinite number of terms (does not terminate) then $f(z)$ has essential singularity at $z = a$

Example

$$f(z) = \sin\left(\frac{1}{z-a}\right)$$

$$= \frac{1}{(z-a)} - \frac{1}{(z-a)^3 \cdot 3!} + \frac{1}{(z-a)^5 \cdot 5!} - \dots$$

Infinite terms of -ve power of $(z-a)$
 $\therefore f(z)$ has essential singularity at $z = a$

pole: If there are finite no. of terms in the principal part of $f(z)$ i.e.

$$b_1 (z-a)^{-1} + b_2 (z-a)^{-2} + \dots + b_m (z-a)^{-m}$$

Then $z = a$ is said to be pole of order m when $m \geq 2$ (First order pole)

$$m \geq 2 \quad \left(\begin{array}{ccc} 2^{nd} & 11 & 11 \end{array} \right)$$

$$\text{Ex: } f(z) = \frac{1}{(z-a)} + \frac{3}{(z-a)^2} + \frac{5}{(z-a)^3}$$

$f(z)$ has 3rd order pole at $z = a$

Zeros of a function

If $f(z)$ be analytic in D then point $z=a$ is said to be zero of $f(z)$ if $f(a)=0$.

A zero of $f(z)$ has order n , if and only if f but also the derivatives.

$f', f'', \dots, f^{n-1}$ are zero but $f^{(n)}(a) \neq 0$

$$f(z) = \cos z \text{ at } z = \pi/2$$

Here $f(\pi/2) = \cos \pi/2 = 0$ so, $f(z)$ has zero at $z = \pi/2$

~~$$f(z) = \cos^2 z \text{ at } z =$$~~

$$f(z) = \sin^2 z \text{ at } z = 0, \pm \pi, \pm 2\pi, \dots$$

$$f(0) = 0$$

$$f''(0) = 1 \neq 0 \text{ (so, } f(z) \text{ has 2nd order}$$

$$f'(0) = 0$$

zero at $z = 0$)

$$f(z) = \sin^2 z \text{ at } z = \pi$$

$$f(\pi) = \dots = 0$$

$$f'(\pi) = 0$$

$$f''(\pi) = -2 \neq 0$$

$\therefore f(z)$ has 2nd order zero at $z = \pi$

$$f(z) = (1 - \cos z)^2 \text{ at } z = 0, z = \pm 2\pi$$



4th order zero

q) $F(z) = \tan z$
location and order of zero

$$F(z) = \tan z = \tan 0, \tan(\pm \pi) \dots$$

$\therefore z = n\pi$ is zero

where $n = \pm 1, \pm 2, \dots$ simple order.

c) $F(z) = (z^2 + 1)(e^z - 1)$

$$F(z) = 0$$

$$z^2 + 1 = 0$$

$$z = \pm i$$

$$e^z = 1 = e^0$$

$$\therefore z = 0$$

g) singularity: $f(z) = z^2 - \frac{1}{z^2}$

$$= \frac{z^4 - 1}{z^2}$$

$$\left. \begin{array}{l} \text{For } z^2 = 0 \\ \text{or, } z^2 = \infty \end{array} \right\} f(z) \text{ not defined}$$

$\therefore z = 0, \infty$ are singular points.

vp) $f(z) = (z - \pi i)^{-2} \cdot \sinh z$ (singularity of type)

$$= \frac{\sinh z}{(z - \pi i)^2}, \quad z - \pi i = 0$$

$$\Rightarrow z = \pi i$$

Imp

Residue, methods of finding residues, Cauchy Residue theorem (statement only) and Evaluation of some integrals using Residue technique.

We have Laurent's Expansion $f(z) =$

$$\sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=1}^{\infty} b_n (z-a)^{-n} \text{ is valid}$$

In the region $r_1 < |z-a| < r_2$

The coeff b_1 of $\frac{1}{z-a}$ is said to be the residue of $f(z)$ at $z=a$.

Note: Residue of $f(z)$ at $z=a$ is $b_1 = \frac{1}{2\pi i} \oint_C f(z) dz$ where C is any simple closed path such that $f(z)$ is analytic in C except at $z=a$.

Formula for finding Residue

(i) If $f(z)$ has simple pole at $z=z_0$ then residue of $f(z)$ at $z=z_0$ is $b = \lim_{z \rightarrow z_0} (z-z_0) f(z)$

ii) If $f(z) = \frac{p(z)}{q(z)}$ residue of $f(z)$ at $z=z_0$ is $b = \frac{p(z_0)}{q'(z_0)}$

iii) If $f(z)$ has n^{th} order pole at $z=z_0$ (if principal part contains finite no. of terms) then Residue at $z=z_0$ is $b = \frac{1}{(n-1)!} \frac{d^{n-1}}{dz^{n-1}} [(z-z_0)^n f(z)]$
i.e. $b = \lim_{z \rightarrow z_0} \frac{1}{(n-1)!} \frac{d^{n-1}}{dz^{n-1}} [(z-z_0)^n f(z)]$

$$\text{i.e. } b = \frac{1}{(m-1)!} \lim_{z \rightarrow z_0} \frac{d^{m-1}}{dz^{m-1}} \left[(z-z_0)^m \cdot f(z) \right]$$

→ Cauchy Residue theorem (CRT): -

Let $f(z)$ be analytic inside and on a simple closed path C and except for finite number of singular points $z_1, z_2, z_3, \dots, z_k$. Then the integral of $f(z)$ C.C.W. around C is equal to $2\pi i$ times the sum of residues at those singular points i.e.

$$\oint_C f(z) dz = 2\pi i \left(\text{sum of residues at poles of } z_1, z_2, z_3, \dots, z_n \right)$$

$$= 2\pi i \sum_{j=1}^n \text{Res. } f(z)_{z=z_j}$$

Ex 6.2

(1). Find the residues at singular points.

9) $f(z) = \frac{4z}{1+z^2}$

here: For singular points $1+z^2=0$

$$z^2 - i^2 = 0$$

$$\text{or } (z+i)(z-i) = 0$$

$\therefore z = \pm i$ are singular pts (poles)

So, $f(z)$ has simple pole at $z=i$ and $z=-i$

Then residues at $z=i$

$$b_1 = \lim_{z \rightarrow i} (z-i) f(z) = \lim_{z \rightarrow i} (z-i) \frac{4z}{(z+i)(z-i)}$$

$$= \frac{4p}{p+1} = 2$$

The Residue at $z = -p$

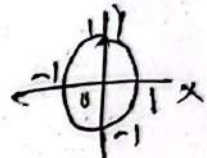
$$= \lim_{z \rightarrow -p} (z+p) F(z) = \lim_{z \rightarrow -p} \frac{(z+p) (4z)}{(z+1)(z-1)}$$

$$= \frac{4(-p)}{-1-p} = \frac{-4p}{-2p} = 2$$

2) Evaluate the following integrals (CCW)

$\oint_C \tan \pi z \, dz$ around $C: |z|=1$ using Cauchy Residue theorem.

soln



$$\text{Here } f(z) = \tan \pi z = \frac{\cos \pi z}{\sin \pi z} \cdot \frac{\sin \pi z}{\cos \pi z}$$

For singular / pole points: $\cos \pi z = 0 = \cos\left(\frac{n\pi}{2}\right)$

$$\therefore \pi z = \pm n \frac{\pi}{2}$$

$$\therefore z = \pm \frac{n}{2} \Rightarrow \pm \frac{1}{2}, \pm 1, \pm \frac{3}{2}, \dots$$

only $z = \pm \frac{1}{2}$ and $z = \pm \frac{3}{2}$ are inside $|z|=1$

Then residue of $f(z)$ at $z = \frac{1}{2}$

$$\therefore \text{Res } f(z) = \lim_{z \rightarrow \frac{1}{2}} \frac{\sin \pi \cdot \frac{1}{2}}{-\pi \sin \pi z}$$

$$= -\frac{2}{\pi}$$

$$\begin{aligned} f(z) &= \frac{p(z)}{q(z)} \\ \text{Res } &= \frac{p(z_0)}{q'(z_0)} \\ q'(z) &= -\pi \sin \pi z \end{aligned}$$

Similarly

$$\text{Res}_{z=-2/2} F(z) = \frac{\sin(-\pi/2)}{-\pi \sin(-\pi/2)}$$

$$= \frac{-1}{\pi}$$

Then,

$$\oint_C \tan \pi z dz = 2\pi i \left(-\frac{1}{\pi} - \frac{1}{\pi} \right)$$

$$= 2\pi i \left(-\frac{2}{\pi} \right) = -4i$$

3) $\oint_C \frac{z+1}{z^4-2z^3} dz$ where $C: |z| = \frac{1}{2}$ using C.R.T
So

$$\text{here } f(z) = \frac{z+1}{z^4-2z^3} = \frac{z+1}{z^3(z-2)}$$

For singular / pole point: $z^3(z-2) = 0$

$$z = 0 \quad z = 2$$

$z = 0$ is 3rd order pole and $z = 2$ is simple pole. lies outside C

$$\text{Residue } F(z) = \frac{1}{(3-1)!} \lim_{z \rightarrow 0} \frac{d^{3-1}}{dz^{3-1}} \left[\frac{z^3(z+1)}{z^3(z-2)} \right]$$

$$= \frac{1}{2!} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} \left[\frac{z+1}{z-2} \right]$$

$$= \frac{1}{2} \lim_{z \rightarrow 0} \left(\frac{6}{(z-2)^3} \right) = \frac{1}{2} \times \frac{6}{-8} = -\frac{3}{8}$$

$$\oint_C \frac{z+1}{z^4-2z^3} dz = 2\pi i \left(-\frac{1}{8}\right)$$

$$= \frac{-3\pi i}{4}$$

$$\frac{d}{dz} \left(\frac{z+1}{(z-2)^2} \right) = \frac{(z-2) \cdot 1 - (z+1) \cdot 2}{(z-2)^3}$$

$$= \frac{-3}{(z-2)^2}$$

$$\therefore \frac{d^2}{dz^2} \left(\frac{z+1}{(z-2)^2} \right) = \frac{6}{(z-2)^3}$$

Imp Ex 6.2

2) d. Evaluate $\oint_C \frac{z \cosh \pi z}{z^4 + 13z^2 + 36} dz$, $C: |z| = \pi$.

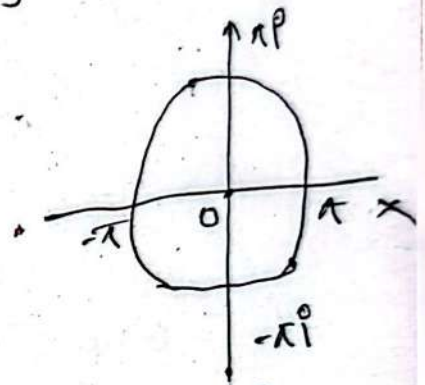
For singular points or poles

$$(z^2)^2 + 9z^2 + 4z^2 + 36 = 0$$

or, $z^2(z^2 + 9) + 4(z^2 + 9) = 0$

or, $(z^2 + 9)(z^2 + 4) = 0$

or, $z^2 = \pm 3i$ and $z^2 = \pm 2i$



here, $z = \pm 2i$ and $z = \pm 3i$ are all simple poles lying inside $|z| = \pi$

Residue $F(z) = \frac{z \cosh \pi z}{4z^3 + 26z}$ at $z = 3i$

Note:-

$$\cosh \pi 3i = \cosh 3\pi i$$

$$\cosh 3\pi i = 1$$

$$= \frac{3i \cosh \pi 3i}{4(3i)^3 + 26(3i)} = \frac{3i \cosh 3\pi i}{-108i + 78i}$$

$$= \frac{-3i \cdot 1}{-30i} = \frac{1}{10}$$

W

Similarly

$$\begin{aligned} \text{Res } f(z) &= \frac{z \cosh \pi z}{4z^3 + 26z} \quad \text{at } z = -3i \\ z = -3i & \\ &= \frac{1}{10} \end{aligned}$$

$$\begin{aligned} \text{Res } f(z) &= \frac{z \cosh \pi z}{4z^3 + 26z} \quad \text{at } z = 2i \\ z = 2i & \end{aligned}$$

$$= \frac{1}{4(2i)^2 + 26} = \frac{1}{10}$$

$$\boxed{\because \cosh 2\pi i = 1}$$

Similarly

$$\begin{aligned} \text{Res } f(z) &= \frac{z \cosh \pi z}{4z^3 + 26z} \quad \text{at } z = -2i \\ z = -2i & \end{aligned}$$

$$= \frac{1}{10}$$

$$\oint_C \frac{z \cosh \pi z}{z^4 + 7z^2 + 36} dz = 2\pi i \left(\frac{1}{10} + \frac{1}{10} + \frac{1}{10} + \frac{1}{10} \right)$$

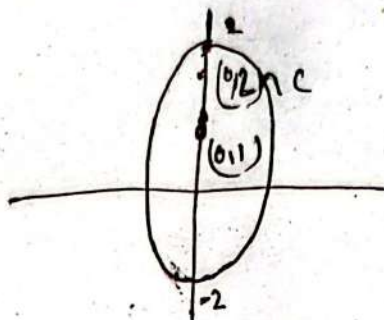
$$= \frac{4\pi i}{5}$$

(3) $\oint \frac{dz}{(z^2+4)^3}$ where $C: |z-p|=2$

here, $z^2+4=0$

$\therefore z = \pm 2i$ are poles of 3rd order

path $|z-p|=2$ centre $(0,1)$ and radius $=2$



$$\begin{aligned} |x+iy-p| &= 2 \\ (x-0)^2 + (y-1)^2 &= 2^2 \end{aligned}$$

only $z = 2i$ lies but $z = -2i$ does not.
we need to find residue of $f(z)$ at $z = 2i$ (3rd order pole)

Residue $b_1 = \frac{1}{(3-1)!} \lim_{z \rightarrow 2i} \left[\frac{d^2}{dz^2} (z-2i)^3 \cdot \frac{1}{(z-2i)^3(z+2i)} \right]$

$$= \frac{1}{2} \lim_{z \rightarrow 2i} \left(\frac{12}{(z+2i)^5} \right)$$

$$= \frac{1}{2} \lim_{z=2i} \left(\frac{12}{(2i+2i)^5} \right)$$

$$= \frac{1}{2} \times \frac{12}{(4i)^5}$$

$$= \frac{-3i}{512}$$

$$\oint \frac{dz}{(z^2+4)^2} = 2\pi i \times \left(\frac{-3i}{512} \right)$$

$$= \frac{3\pi}{256}$$

$$\begin{aligned} i^2 &= (-1) \\ i &= \sqrt{-1} \end{aligned}$$

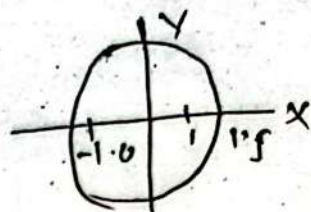
$$\begin{aligned} i^2 &= (-1) \\ i &= \sqrt{-1} \end{aligned}$$

$$\begin{aligned} i^2 &= (-1) \\ i &= \sqrt{-1} \end{aligned}$$

$$\begin{aligned} i^2 &= (-1) \\ i &= \sqrt{-1} \end{aligned}$$

9. $\oint \frac{\tan z}{z^2 - 1} dz$ where $C: |z| = 3/2$

For poles $z = \pm 1$



$z = \pm 1$ both lying in $|z| = 3/2$ and both are

$\therefore \text{Res } f(z)_{z=1} = \frac{\tan z}{2z} \text{ at } z=1$

$= \frac{\tan 1}{2}$

$\text{Res } f(z)_{z=-1} = \dots = \frac{+\tan 1}{+2} = \frac{\tan 1}{2}$

$\therefore \oint \frac{\tan z}{z^2 - 1} dz = 2\pi i \left(\frac{\tan 1}{2} + \frac{\tan 1}{2} \right)$

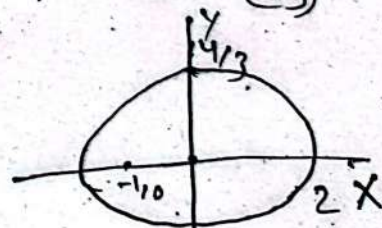
$= 2\pi i \tan 1$

7). Evaluate $\int \frac{e^{2z}}{(z+1)^3} dz$ where $C: 4x^2 + 9y^2 = 16$

$\Rightarrow \frac{x^2}{2^2} + \frac{y^2}{(4/3)^2} = 1$

$z = -1$ lies inside C

$z = -1$ is 3rd order pole.



$\therefore \text{Res } f(z) \text{ at } z = -1$

$b = \frac{1}{2!} \lim_{z \rightarrow -1} \left[\frac{d^2}{dz^2} \left[(z+1)^3 \cdot \frac{e^{2z}}{(z+1)^3} \right] \right]$

$$b = \lim_{z \rightarrow -1} (4e^{2z})$$

$$= 4e^{-2} = \frac{2}{e^2}$$

$$\oint_C \frac{e^{2z}}{(z+1)^3} dz = 2\pi i \times \frac{2}{e^2}$$

$$= \frac{4\pi i}{e^2}$$

u

$$\begin{array}{l} x^2 \\ 2x \\ 2 \\ 0 \end{array} \quad \frac{\sin 3x}{3}$$

Unit 7

FOURIER Integrals and FOURIER transforms

pre-knowledge

Integration by parts.

$$\int_U F(u) \frac{d}{du} g(u) du = U g_1 - U g_2 + U'' g_3 - U'' g_4 + \dots$$

$$U = F(u)$$

$$U' = F'(u)$$

$$U'' = F''(u)$$

$$u_1 = \int g(u) du$$

$$u_2 = \int u_1 du$$

$$u_3 = \int u_2 du \dots$$

"I LATE" rule to choose U and u

D I method for integration by parts.

$$\int \overset{D}{x^2} \overset{I}{\cos 3x} dx$$

	+	-	+	-
D	x^2	$2x$	2	0
I	$\frac{\sin 3x}{3}$	$-\frac{\cos 3x}{9}$	$-\frac{\sin 3x}{27}$	$\frac{\cos 3x}{31}$

$$\begin{array}{r} D \\ x^2 \\ 2x \\ 2 \\ 0 \end{array}$$

$$\begin{array}{r} I \\ \frac{\sin 3x}{3} \\ -\frac{\cos 3x}{9} \\ -\frac{\sin 3x}{27} \\ \frac{\cos 3x}{31} \end{array}$$

$$\begin{array}{r} D \\ x^2 \\ 2x \\ 2 \\ 0 \end{array} \quad \begin{array}{r} I \\ \cos 3x \\ \frac{\sin 3x}{3} \\ -\frac{\cos 3x}{9} \\ -\frac{\sin 3x}{27} \end{array}$$

Fourier series

Let $f(x)$ be a function of x , having period $2L$

Then Fourier series of $f(x)$ is

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

$$\text{where, } a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx$$

Note

If $f(x)$ has period 2π then the Fourier series becomes,

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$\text{where } a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$$

} $n = 1, 2, 3, \dots$

* Obtain complex Form of Fourier series from the real form;

We have the real form of Fourier series of $f(x)$ of period 2π is

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \left(\frac{e^{inx} + e^{-inx}}{2} \right) + b_n \left(\frac{e^{inx} - e^{-inx}}{2i} \right) \right]$$

$$= a_0 + \sum_{n=1}^{\infty} \left(\frac{a_n - ib_n}{2} \right) e^{inx} + \sum_{n=1}^{\infty} \left(\frac{a_n + ib_n}{2} \right) e^{-inx}$$

Here, $(a_n - ib_n)$ and $(a_n + ib_n)$ are complex conjugates of each other and so are e^{inx} and e^{-inx}

Re-indexing this sum

$$f(x) = a_0 + \underbrace{\sum_{n=1}^{\infty} \left(\frac{a_n - ib_n}{2} \right) e^{inx}}_{c_n} + \underbrace{\sum_{n=1}^{\infty} \left(\frac{a_{-n} + ib_{-n}}{2} \right) e^{-inx}}_{c_{-n}}$$

$\downarrow$ c_0 $\downarrow$ c_n $\downarrow$ c_{-n}

$$\therefore f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx} \text{ is compact form}$$

is the complex form of Fourier series.

where, $c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$

~~$$\frac{a_n + ib_n}{2} = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(x) e^{-inx} dx$$~~

Fourier Integral

The Fourier integral of $f(x)$ is given by

$$f(x) = \int_0^{\infty} [A(\omega) \cos \omega x + B(\omega) \sin \omega x] d\omega$$

$$\text{where, } A(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \cos \omega x dx$$

$$\& B(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \sin \omega x dx$$

Odd Function | even.

If $f(x) = f(-x)$ even

$f(-x) = -f(x)$ odd

even; $x^2, x^4, \cos x \dots$

odd: $x^3, \sin x, \tan x \dots$

Fourier sine series & cosine series of $f(x)$ of period 2π

Full series

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

For $f(x)$ is odd $\therefore a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = 0$

$$\therefore f(x) = \sum_{n=1}^{\infty} b_n \sin nx \quad \text{and } a_n = 0$$

where $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$

For $f(x)$ even $b_n = 0$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx$$

$$\text{where } a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx, \quad a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx.$$

Fourier sine integral

(i) For odd function. $f(x)$

$$f(x) = \int_0^{\infty} b(w) \sin wx dw$$

$$\text{where } b(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \sin wx dx$$

Fourier cosine integral

ii) For $f(x)$ even

$$f(x) = \int_0^{\infty} A(w) \cos wx dw$$

$$\text{where, } A(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \cos wx dx$$

$$\int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx)$$

$$\int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx)$$

Fourier Integral

1)

V-V Imp

$$\text{show that } \int_0^{\infty} \frac{\cos xw + w \sin xw}{1+w^2} dw = \begin{cases} 0 & \text{if } x < 0 \\ \pi/2 & \text{if } x = 0 \\ \pi e^{-x}, & \text{if } x > 0 \end{cases} \quad \text{--- (1)}$$

$$\text{Let } f(x) = \begin{cases} 0 & \text{For } x < 0 \\ \pi/2 & \text{For } x = 0 \\ \pi e^{-x} & \text{For } x > 0 \end{cases} \quad \text{--- (2) we need to show (1) = (2)}$$

The integral (1) contains both cosine and sine function, so we have to find the Fourier integral of $f(x)$.

$$\text{As, } f(x) = \int_0^{\infty} [A(w) \cos wx + B(w) \sin wx] dw \quad \text{--- (3)}$$

$$\text{where, } A(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \cos wx dx$$

$$= \frac{1}{\pi} \left[\int_{-\infty}^0 0 \cdot \cos wx dx + \int_0^0 \pi/2 \cdot \cos wx dx + \int_0^{\infty} \pi e^{-x} \cos wx dx \right]$$

$$= \int_0^{\infty} e^{-x} \cos wx dx.$$

 $\therefore A(w) = \left[\frac{e^{-x}}{1+w^2} (-\cos wx + w \sin wx) \right]_0^{\infty}$

$$\begin{aligned}
 b(\omega) &= \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \sin \omega x \, dx = \frac{1}{\pi} \int_0^{\infty} e^{-x} \sin \omega x \, dx \\
 &= \left[\frac{e^{-x}}{1+\omega^2} (-\sin \omega x - \omega \cos \omega x) \right]_0^{\infty} \\
 &= \frac{\omega}{1+\omega^2} \text{ as } e^{-\infty} \rightarrow 0
 \end{aligned}$$

From eqn (3)

$$f(x) = \int_0^{\infty} \left[\frac{1}{1+\omega^2} \cos \omega x + \left(\frac{\omega}{1+\omega^2} \right) \sin \omega x \right] d\omega$$

$$f(x) = \int_0^{\infty} \left(\frac{\cos \omega x + \omega \sin \omega x}{1+\omega^2} \right) d\omega \quad \dots (4)$$

proved

9) prove that $\int_0^{\infty} \left(\frac{1 - \cos \pi \omega}{\omega} \right) \sin x \, d\omega = \begin{cases} \pi/2 & \text{For } 0 < x < \pi \\ 0 & \text{For } x > \pi \end{cases}$

Complex Form of Fourier Integral and

Fourier transform :

we have real form of Fourier integral of $F(x)$ is

$$F(x) = \int_0^{\infty} [A(w) \cos wx + B(w) \sin wx] dw \quad \dots (1)$$

where $A(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} F(t) \cos wt \, dt$ and

$$B(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} F(t) \sin wt \, dt$$

putting $A(w)$ and $B(w)$ in (1) we get.

$$F(x) = \frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} F(t) [\cos wt \cos wx + \sin wt \sin wx] dt dw$$

$$= \frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} F(t) \cos(wx - wt) dt dw$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(t) \cos(wx - wt) dt dw, \text{ since cosine is even function} \quad \dots (2)$$

but,

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(t) \sin(wx - wt) dt dw = 0 \text{ as sine function is odd.}$$

(2) can be written as $F(x) = \frac{1}{2\pi} \left[\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(t) \cos(wx - wt) dt dw + i \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(t) \sin(wx - wt) dt dw \right]$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t) \cdot e^{i(\omega x - \omega t)} dt d\omega$$

$$\therefore f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t) e^{i\omega(x-t)} dt d\omega \text{ is said to be}$$

Complex Fourier Integral,

Complex Form of Fourier transform and its Inverse

We have complex Fourier integral $f(x) =$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(t) e^{i\omega(x-t)} dt d\omega \quad \text{--- (1)}$$

$$= \frac{1}{\sqrt{2\pi}} \left[\int_{-\infty}^{\infty} \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt \right] e^{i\omega x} d\omega \right]$$

$$f(x) = \frac{1}{\sqrt{2\pi}} f(\omega) \cdot e^{i\omega x} \quad \text{--- (2)}$$

where $f(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$ is said to be

Fourier transform of $f(t)$ or $f(x)$ & from (2),

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(\omega) e^{i\omega x} d\omega \text{ is said to be Inverse}$$

Fourier transform of $f(\omega)$

Linearity of Fourier transform

Let $f(x)$ and $g(x)$ are functions whose Fourier transforms are $\mathcal{F}(f)$ and $\mathcal{F}(g)$ respectively, then for any constants a & b we have,

$$\mathcal{F}[af(x) + bg(x)] = a\mathcal{F}f(x) + b\mathcal{F}g(x)$$

Imp Fourier transform of derivative of $f(x)$

Theorem Let $f(x)$ be continuous on x -axis and $f(x) \rightarrow 0$ as $|x| \rightarrow \infty$ and $f(x)$ is absolutely integrable on x -axis.

Then, $\mathcal{F}[f(x)] = p.w \mathcal{F}[f(x)]$ and $\mathcal{F}[f'(x)] = -i\omega \mathcal{F}[f(x)]$

proof By definition of Fourier transform,

$$\begin{aligned}\mathcal{F}[f'(x)] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f'(x) e^{-i\omega x} dx \\&= \frac{1}{\sqrt{2\pi}} \left[\left\{ e^{-i\omega x} f(x) \right\}_{-\infty}^{\infty} + i\omega \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx \right] \\&= 0 + \frac{1}{\sqrt{2\pi}} i\omega \mathcal{F}[f(x)] \\&= i\omega \mathcal{F}[f(x)] \text{ proved}\end{aligned}$$

similarly $\mathcal{F}$

Exercise 7.3

1) Find Fourier transform of

$$f(x) = \begin{cases} 1, & \text{for } a < x < b \\ 0, & \text{otherwise,} \end{cases}$$

we have, Fourier transform of $f(x)$ is

$$F(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} 1 \cdot e^{-i\omega x} dx$$

$$= \frac{1}{\sqrt{2\pi}} \left[\int_{-\infty}^0 0 \cdot e^{-i\omega x} dx + \int_0^a 0 \cdot e^{-i\omega x} dx + \int_a^b 1 \cdot e^{-i\omega x} dx + \int_b^{\infty} 0 \cdot e^{-i\omega x} dx \right]$$

$$= \frac{1}{\sqrt{2\pi}} \left[\frac{e^{-i\omega x}}{-i\omega} \right]_a^b = \frac{1}{\sqrt{2\pi}} \frac{1}{(-i\omega)} (e^{-i\omega b} - e^{-i\omega a})$$

$$= \frac{1}{\omega\sqrt{2\pi}} (e^{-i\omega b} - e^{-i\omega a})$$

4) $f(x) = \begin{cases} x e^{-x} & \text{if } x > 0 \\ 0 & \text{if } x < 0 \end{cases}$

we have, Fourier transform of $f(x)$ is

$$F(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_0^{\infty} x e^{-x} \cdot e^{-i\omega x} dx$$

$$\begin{aligned}
 &= \frac{1}{\sqrt{2\pi}} \int_0^{\infty} x \cdot e^{-(1+i\omega)x} dx \\
 &= \frac{1}{\sqrt{2\pi}} \left[-x(1+i\omega) e^{-x(1+i\omega)} \right]_0^{\infty} - \frac{1}{\sqrt{2\pi}} \left[\frac{e^{-x(1+i\omega)}}{(1+i\omega)^2} \right]_0^{\infty} \\
 &= 0 + \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{(1+i\omega)^2} \quad \underline{\underline{\text{Ans}}}
 \end{aligned}$$

7) $f(x) = e^{-x^2/2}$

$$\begin{aligned}
 \underline{\underline{\text{V.I.M.P}}}\quad \mathcal{F}[f(x)] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-x^2/2} \cdot e^{-i\omega x} dx \\
 &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x^2/2 + i\omega x)} dx \\
 &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left(\frac{x^2 + 2i\omega x}{2}\right)} dx \\
 &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x^2 + 2i\omega x + (i\omega)^2 - (i\omega)^2)} dx \\
 &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left(\frac{x+i\omega}{\sqrt{2}}\right)^2} \cdot e^{-\omega^2/2} dx \\
 &= \frac{1}{\sqrt{2\pi}} e^{-\omega^2/2} \int_{-\infty}^{\infty} e^{-\left(\frac{x+i\omega}{\sqrt{2}}\right)^2} dx
 \end{aligned}$$

$$\text{let, } \frac{x+iw}{\sqrt{2}} = y$$

$$\therefore x+iw = \sqrt{2}y$$

$$1+0 = \sqrt{2} \frac{dy}{dx}$$

$$\therefore dx = \sqrt{2} dy$$

$$\text{Formula, } \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

$$= \frac{1}{\sqrt{2\pi}} \cdot e^{-w^2/2} \cdot \sqrt{2} \int_{-\infty}^{\infty} e^{-y^2} dy$$

$$= \frac{1}{\sqrt{2\pi}} \cdot e^{-w^2/2} \cdot \sqrt{\pi}$$

$$= e^{-w^2/2}$$

define convolution of two function

→ The convolution of two function $f(x)$ and $g(x)$ be denoted by $f * g$ and is defined by

$$f * g = \int_{-\infty}^{\infty} f(p) \cdot g(x-p) dx = \int_{-\infty}^{\infty} f(x-p) g(p) dp.$$

soin.

Let $f(x)$ and $g(x)$ be piecewise continuous, bounded and absolutely integrable on x -axis then

$$\mathcal{F}[f * g] = \sqrt{2\pi} \mathcal{F}(f) \mathcal{F}(g)$$

→ By definition of convolution of f and g we have

$$f * g = \int_{-\infty}^{\infty} f(p) \cdot g(x-p) \cdot dp \quad \dots (1)$$

Taking Fourier transform on both sides of (1)

$$\mathcal{F}[f * g] = \mathcal{F}\left[\int_{-\infty}^{\infty} f(p) g(x-p) dp\right]$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(p) g(x-p) dp \right] e^{-i\omega x} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(p) g(x-p) e^{-i\omega x} dp \cdot dx$$

By the order of integrating, let $x-p = q \Rightarrow x = p+q$
 $dx = dq$

$$\text{As } x \rightarrow -\infty \quad q \rightarrow -\infty$$

$$x \rightarrow \infty \quad q \rightarrow \infty$$

From (1)

$$\mathcal{F}[f * g] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(p) \cdot g(q) \cdot e^{-i\omega(p+q)} dq \cdot dp$$

$$= \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(p) e^{-i\omega p} dp \right] \cdot \left[\sqrt{2\pi} \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(q) e^{-i\omega q} dq \right]$$

$$= \sqrt{2\pi} \mathcal{F}(p) \cdot \mathcal{F}(q) \quad \underline{\text{proved}}$$

* Parseval's Identity for Fourier Transform

Let $F(x)$ and $g(x)$ be piecewise, continuous, bounded, on x -axis, with Fourier transforms $\mathcal{F}(F)$ and $\mathcal{F}(g)$ respectively.

Then, $\int_{-\infty}^{\infty} F(w) \cdot \overline{g(w)} dw = \int_{-\infty}^{\infty} F(x) \cdot \overline{g(x)} dx$ where bar (—) denote the conjugates of respective functions.

Also

$$\int_{-\infty}^{\infty} |F(w)|^2 dw = \int_{-\infty}^{\infty} |F(x)|^2 dx \quad \boxed{\because |F(w)|^2 = |\overline{F(x)}|^2}$$

Parseval's identity for Fourier sine transform

$$\int_0^{\infty} F_s(w) \cdot h_s(w) dw = \int_0^{\infty} F(x) \cdot g(x) dx$$

and $\int_0^{\infty} |F_s(w)|^2 dw = \int_0^{\infty} |F(x)|^2 dx$

Ex. 11 Imp Find Fourier transform of

$$9) \quad F(x) = \begin{cases} x & \text{if } |x| < 9 \Rightarrow -9 < x < 9 \\ 0 & \text{if } |x| > 9 \end{cases}$$

Fourier transform of $F(x)$ is given by

$$\mathcal{F}[F(x)] = \frac{1}{(2\pi)^{1/2}} \int_{-9}^9 x \cdot e^{-iwx} dx \quad \underline{\text{Hence}}$$

$$= \frac{\sqrt{2}}{\sqrt{\pi}} [i(a w \cos wq - \sin wq)] \quad \underline{\text{Ans}}$$

2. Amp

10) Find Fourier transform of $f(x) = \begin{cases} 1-x^2 & \text{For } |x| < 1 \\ 0 & \text{For } |x| > 1 \end{cases}$ and then show that.

$$\int_0^{\infty} \frac{x(\cos x - \sin x)}{x^3} \cos \frac{x}{2} dx = \frac{-3\pi}{16}$$

We have Fourier transform of $f(x)$ is

$$\mathcal{F}[f(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-1}^1 (1-x^2) e^{-i\omega x} dx$$

$$= \frac{1}{\sqrt{2\pi}} \left[\frac{(1-x^2)e^{-i\omega x}}{(-i\omega)} - \frac{(2x)e^{-i\omega x}}{(-i\omega)^2} + \frac{(-2)e^{-i\omega x}}{(-i\omega)^3} \right]_{-1}^1$$

$$= \frac{1}{\sqrt{2\pi}} \left[0 + \frac{2e^{-i\omega} + 2e^{i\omega}}{(-i\omega)^2} + \frac{2}{-i\omega^3} (e^{-i\omega} - e^{i\omega}) \right]$$

$$= \frac{1}{\sqrt{2\pi}} \left[\frac{-4}{\omega^2} \cos \omega - \frac{4}{\omega^3} \sin \omega \right]$$

$$= \frac{-4}{\sqrt{2\pi}} \left[\frac{\omega \cos \omega - \sin \omega}{\omega^3} \right] \text{ is the req Fourier.}$$

For the 2nd part.

We have the inverse Fourier transform of $\mathcal{F}[f(x)]$

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \mathcal{F}[f(x)] \cdot e^{i\omega x} d\omega$$

$$\frac{-4}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\frac{\omega \cos \omega - \sin \omega}{\omega^3} \right) \cdot (\cos \omega x + i \sin \omega x) d\omega = \sqrt{2\pi} \cdot f(x)$$

$$= \sqrt{2\pi} \begin{cases} 1-x^2 & \text{For } |x| < 1 \\ 0 & \text{For } |x| > 1 \end{cases}$$

Since, $\frac{w \cos w}{w^3}$ is an even function, ~~$\cos w$~~

$\left(\frac{w \cos w - \sin w}{w^3} \right) \cdot \sin wx$ is odd function then,

$$\int_{-\infty}^{\infty} \frac{w \cos w - \sin w}{w^3} \sin wx \, dw = 0 \text{ and}$$

$\left(\frac{w \cos w - \sin w}{w^3} \right) \cos wx$ is even.

$$\text{and } \int_{-\infty}^{\infty} \left(\frac{w \cos w - \sin w}{w^3} \right) \cos wx \, dw = 2 \int_0^{\infty} \left(\frac{w \cos w - \sin w}{w^3} \right) \cos wx \, dw$$

ii) becomes

$$\frac{-\sqrt{4}}{\sqrt{4}\pi} \cdot 2 \int_0^{\infty} \frac{w(\cos w - \sin w)}{w^3} \cdot \cos xw \, dw = \frac{-\pi}{4} (1-x^2)$$

setting $x = \frac{1}{2}$ and $w = x$

$$\int_0^{\infty} \frac{w(\cos w - \sin w)}{w^3} \cdot \cos \frac{x}{2} \, dw$$

$$= \int_0^{\infty} \frac{(x \cos x - \sin x)}{x^3} \cdot \cos \left(\frac{x}{2} \right) \, dx = \frac{-3\pi}{16}$$

~~~~~



Imp

## Fourier transforms

Ex: 7.3

cosine

- (9) Find the Fourier transform of  $f(x) = e^{-mx}$  for  $m > 0$  and  $m \in \mathbb{R}$ . Show that  $\int_0^{\infty} \frac{\cos kx}{1+x^2} dx = \frac{\pi}{2} e^{-k}$ .

soln, cosine

Fourier transform of  $f(x)$  is given by

$$\mathcal{F}[f(x)] = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} f(x) \cos wx dx$$

$$\therefore \mathcal{F}[f(x)] = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-mx} \cos wx dx$$

$$= \frac{2}{\sqrt{2\pi}} \left[ \frac{e^{-mx}}{m^2 + w^2} (-m \cos wx + w \sin wx) \right]_0^{\infty}$$

$$= \sqrt{\frac{2}{\pi}} \left( 0 + \frac{mw}{m^2 + w^2} \right)$$

$$\mathcal{F}[f(x)] = \sqrt{\frac{2}{\pi}} \frac{mw}{(m^2 + w^2)}$$
 is the reqd Fourier transform of  $f(x)$

we have Inverse Fourier transform of  $\mathcal{F}[f(x)]$  is

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \mathcal{F}[f(x)] \cdot e^{iwx} dw \cos wx dw$$

$$f(x) = \frac{2}{\sqrt{2\pi}} \int_0^{\infty} \sqrt{\frac{2}{\pi}} \left( \frac{m}{m^2 + w^2} \right) e^{iwx} dw \cos wx dw$$

$$\text{or, } \frac{\pi}{2} f(x) = \int_0^{\infty} \frac{m}{m^2 + w^2} \cdot e^{iwx} dw \cos wx dw$$

$$\text{or, set } w = k \text{ and } w = x, m = 1$$

$$\text{or } \int_0^{\infty} \frac{\cos kx}{1+x^2} dx = \frac{\pi}{2} e^{-k}$$



## Fourier transforms

3) Find Fourier sine transform of  $f(x) = \frac{e^{-qx}}{x}$  then, show that  $\int_0^\infty \tan^{-1}\left(\frac{x}{q}\right) \sin x dx = \frac{\pi}{2} e^{-q}$

we have Fourier sine transform of  $f(x)$  is  $\mathcal{F}_s[f(x)]$

$$= \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cdot \sin wx dx.$$

$$\therefore \mathcal{F}_s[f(x)] = \sqrt{\frac{2}{\pi}} \int_0^\infty \frac{e^{-qx}}{x} \sin wx dx = I \text{ say,}$$

$$\frac{dI}{dw} = \sqrt{\frac{2}{\pi}} \int_0^\infty \frac{e^{-qx}}{x} \cdot (\cos wx) x \cdot dx$$

$$\therefore \frac{dI}{dw} = \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-qx} \cos wx dx$$

Integrate - - - -

$$I = \sqrt{\frac{2}{\pi}} \left[ \frac{e^{-qx}}{q^2 + w^2} (-q \cos wx + w \sin wx) \right]_0^\infty$$

$$\therefore \mathcal{F}_s[f(x)] = \sqrt{\frac{2}{\pi}} \left( \frac{q}{q^2 + w^2} \right) \because e^{-\infty} \rightarrow 0$$

is the Fourier sine transform of  $f(x) =$

$$\mathcal{F}_s[f(x)] = \sqrt{\frac{2}{\pi}} \tan^{-1}\left(\frac{w}{q}\right) \quad \underline{\underline{=}}$$

taking inverse Fourier sine transform

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty \sqrt{\frac{2}{\pi}} \tan^{-1}\left(\frac{w}{q}\right) \cdot \sin wx dw$$

$$\frac{\pi}{2} f(x) = \int_0^\infty \tan^{-1}\left(\frac{w}{q}\right) \cdot \sin wx dw.$$

$$\text{or, } \frac{\pi}{2} \cdot \frac{e^{-qx}}{x} = \int_0^\infty \tan^{-1}\left(\frac{w}{q}\right) \sin wx dw$$



$$\text{set } x = w = 1$$

$$\therefore \int_0^{\infty} \tan^{-1}\left(\frac{x}{q}\right) \sin x dx = \frac{\pi}{2} e^{-q} \quad \underline{\underline{\text{proved}}}$$

Imp. 24 find fourier sine transform of  $e^{-x}$  then show

$$\text{that } \int_0^{\infty} \frac{dx}{(1+x^2)^2} = \frac{\pi}{4}$$

$$\therefore \mathcal{F}_s[f(x)] = \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-x} \sin wx dx.$$

$$= \sqrt{\frac{2}{\pi}} \left[ \frac{e^{-x}}{1+w^2} (-\cos wx - w \sin wx) \right]_0^{\infty}$$

$$= \sqrt{\frac{2}{\pi}} \left( \frac{1}{1+w^2} \right) \text{ as } e^{-\infty} \rightarrow 0$$

$\therefore \mathcal{F}_s[f(x)] = \sqrt{\frac{2}{\pi}} \left( \frac{1}{1+w^2} \right)$  is the required fourier sine transform of  $f(x)$ .

now using parseval's identity,

squaring both sides

$$\int_0^{\infty} |f_s(w)|^2 dw = \frac{2}{\pi} \int_0^{\infty} \frac{1}{(1+w^2)^2}$$

$$= \int_0^{\infty} (e^{-x})^2 dx = \frac{2}{\pi} \int_0^{\infty} \frac{1}{(1+w^2)^2} dw$$

$$\Rightarrow \int_0^{\infty} e^{-2x} dx = \frac{2}{\pi} \int_0^{\infty} \frac{1}{(1+w^2)^2} dw$$

$$\Rightarrow \left[ \frac{e^{-2x}}{-2} \right]_0^{\infty} = \frac{2}{\pi} \int_0^{\infty} \frac{1}{(1+w^2)^2} dw$$

$$\Rightarrow \frac{1}{2} = \frac{2}{\pi} \int_0^{\infty} \frac{1}{(1+w^2)^2} dw$$

$$\int_0^{\infty} \frac{1}{(1+w^2)^2} dw = \pi/4$$

$w \rightarrow x$  replacing

$$\int_0^{\infty} \frac{1}{(1+x^2)^2} = \pi/4$$

proved

Example solve the integral eqn.

$$\int_0^{\infty} f(t) \cos wt dt = \begin{cases} 1-w & \text{for } 0 \leq w \leq 1 \\ 0 & \text{for } w > 1 \end{cases}$$

we have Fourier cosine transform of

$f(x)$  is

$$F_0 = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t) \cos wt dt \quad \text{--- (1) here } f_0$$

by inverse Fourier cosine transform,  $f_0 = \begin{cases} 1-w & \text{for } 0 \leq w \leq 1 \\ 0 & \text{for } w > 1 \end{cases}$

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f_c \cdot \cos wx dw$$

$$= \sqrt{\frac{2}{\pi}} \int_0^1 (1-w) \cos wx dw \quad \text{---}$$

$$\Rightarrow f(x) = \frac{2}{\pi} \left( \frac{1 - \cos x}{x^2} \right)$$



## Chapter - 9

### The z-transform and its applications

\* **Definition** :- let  $f(t)$  be a function of time period  $T$ , where  $t$  is non-negative and defined by  $t = 0T, 1T, 2T, \dots$  which are discrete values of  $t$ . then z-transform of  $f(t)$  denoted by  $Z[f(t)]$  and is defined by  $Z[f(t)] = \sum_{n=0}^{\infty} f(nT) z^{-n}$  which is a function of  $z$  and converges.

(Here fixed positive no.  $T$  is referred to as sampling time period)

#### Application

z-transform converts difference Equations of discrete time system to algebraic eqn. which simplifies the discrete time system analysis.

(Laplace and z-transforms are common except that Laplace transform deals with continuous time signals and system but z-transform deals with discrete time signals.)

#### uses

- digital signal processing
- Filter design
- system analysis
- solving difference eqn
- stability analysis.

#### Important expansion

$$(1) \quad e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$(2) \quad \log_e(1+x) = \frac{x}{1} - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n}$$



$$(3) \log_e(1-x) = \frac{-x}{1} - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \dots = - \sum_{n=1}^{\infty} \frac{x^n}{n}$$

$$(4) \frac{a}{1-r} = a + ar + ar^2 + \dots \quad |r| < 1$$

$$(5) (1+x)^n = 1 + \frac{nx}{1!} + \frac{n(n-1)}{2!} x^2 + \dots \quad |x| < 1$$

$$(6) \frac{1}{1-x} = 1 + x + x^2 + \dots \quad \infty$$

$$(7) \frac{1}{(1-x)^2} = 1 + 2x + 3x^2 + 4x^3 + \dots \quad \infty$$

Linearity of z-transforms.

Let  $f(t)$  and  $g(t)$  are two functions of  $t$  ( $t = nT$ )

for  $a, b$  constants.

$$\text{Then, } z[a f(t) + b g(t)] = a z[f(t)] + b z[g(t)]$$

proof by defn

$$z[a f(t) + b g(t)] = \sum_{n=0}^{\infty} [a f(t) + b g(t)] z^{-n}$$

$$= \sum_{n=0}^{\infty} a f(t) z^{-n} + \sum_{n=0}^{\infty} b g(t) z^{-n}$$

$$= a \cdot z[f(t)] + b z[g(t)]$$



① Find z-transform of  $f(t) = 1$   $z \rightarrow (ap + a)$   
 $z \rightarrow smay$

Then  $z[F(t)] = \sum_{n=0}^{\infty} 1 \cdot z^{-n}$

$$= 1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots \dots \infty$$

$$z(1) = \frac{1}{1 - \frac{1}{z}} = \frac{z}{z-1} \quad \sum$$

z-transform of  $(-1)^n$

$$z[(-1)^n] = \sum_{n=0}^{\infty} (-1)^n \cdot z^{-n}$$

$$= 1 - \frac{1}{z} + \frac{1}{z^2} - \frac{1}{z^3} + \dots$$

$$= \frac{1}{\left(1 + \frac{1}{z}\right)} = \frac{z}{z+1}$$

z-transform of  $\frac{1}{n!}$

$$z\left[\frac{1}{n!}\right] = \sum_{n=0}^{\infty} \left(\frac{1}{n!}\right) z^{-n}$$

$$= 1 + \frac{(1/z)}{1!} + \frac{\left(\frac{1}{z}\right)^2}{2!} + \frac{(1/z)^3}{3!} + \dots = e^{1/z}$$

$$z \left[ \frac{1}{n+1} \right] = ?$$

$$\therefore z \left[ \frac{1}{n+1} \right] = \sum_{n=0}^{\infty} \frac{1}{n+1} \cdot z^{-n}$$

$$= 1 + \frac{1}{2} \cdot \frac{1}{z} + \frac{1}{3} \cdot \frac{1}{z^2} + \frac{1}{4} \cdot \frac{1}{z^3} + \dots \infty$$

$$\log_e(1-x) = -x$$

$$= z \left[ \left( \frac{1}{z} \right) + \frac{\left( \frac{1}{z} \right)^2}{2} + \frac{\left( \frac{1}{z} \right)^3}{3} + \frac{\left( \frac{1}{z} \right)^4}{4} + \dots \infty \right]$$

$$= -z \log_e \left( 1 - \frac{1}{z} \right)$$

$$= z \log_e \left( 1 - \frac{1}{z} \right)^{-1} = z \log_e \left( \frac{z}{z-1} \right)$$

Q1 Find the z-transform of unit ramp function.

We have unit ramp function  $f(t) = \begin{cases} t & \text{for } t \geq 0 \\ 0 & \text{for } t < 0 \end{cases}$

$$\text{Then } z[f(t)] = \sum_{n=0}^{\infty} t z^{-n}$$

$$= \sum_{n=0}^{\infty} nT z^{-n}$$

$$= T \sum_{n=0}^{\infty} \frac{n}{z^n}$$

$$= T \sum_{n=0}^{\infty} \frac{n}{z^n} = T \left[ \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots \right]$$

$$= \frac{T}{2} \left[ 1 + \frac{1}{z} + \frac{1}{z^2} + \dots \right]$$

$$= \frac{T}{2(1-1/z)} = \frac{T}{2} \cdot \frac{z}{z-1}$$



Theorem

$$z[t^k] = -T_z \cdot \frac{d}{dz} [z \cdot t^{(k-1)}]$$

by definition

$$z(t^k) = \sum_{n=0}^{\infty} (nT)^k \cdot z^{-n} \dots \textcircled{1}$$

$$z[t^{k-1}] = \sum_{n=0}^{\infty} (nT)^{k-1} \cdot z^{-n}$$

$$= \sum_{n=0}^{\infty} n^{k-1} \cdot T^{k-1} \cdot z^{-n} \dots \textcircled{2}$$

diff. w.r to  $z$

$$\therefore \frac{d}{dz} [z(t^{k-1})] = \sum_{n=0}^{\infty} n^{k-1} T^{k-1} (-n) z^{-n-1}$$

multiplying by  $(-T_z)$  on both sides,

$$-T_z \cdot \frac{d}{dz} [z(t^{k-1})] = \sum_{n=0}^{\infty} n^k \cdot T^k \cdot z^{-n}$$

$$= \sum_{n=0}^{\infty} (nT)^k \cdot z^{-n} = \sum_{n=0}^{\infty} t^k \cdot z^{-n}$$

$$= z[t^k] \text{ proved}$$

$$\therefore z(t^k) = -T_z \frac{d}{dz} [z(t^{k-1})] \text{ --- } \textcircled{3}$$

put  $k=1$  — (3) eqn

$$z(t) = -Tz \frac{d}{dz} [z(t)]$$

$$= -Tz \frac{d}{dz} \left( \frac{z}{z-1} \right)$$

$$= -Tz \cdot \left\{ -\frac{1}{(z-1)^2} \right\}$$

$$= \frac{TzTz}{(z-1)^2}$$

put  $k=2$  — (3) eqn

$$z(t^2) = -Tz \cdot \frac{d}{dz} [z(t)]$$

$$= -Tz \cdot \frac{d}{dz} \left[ \frac{Tz}{(z-1)^2} \right]$$

$$= -T^2z \left[ \frac{(z-1)^2 \cdot 1 - z \cdot 2(z-1) \cdot 1}{(z-1)^4} \right]$$

$$= \frac{T^2z(z+1)}{(z-1)^3}$$

Σ



## Some theorems on z-transform

Imp

### First shifting theorem of z-transform

State and prove "First shifting theorem of z-transform"

$$\text{If } \mathcal{Z}[f(t)] = F(z) \text{ then } \mathcal{Z}[e^{-at} f(t)] = F(ze^{aT})$$

proof: By definition of z-transform,

$$\begin{aligned} \mathcal{Z}[e^{-at} f(t)] &= \sum_{n=0}^{\infty} e^{-anT} f(nT) z^{-n} = \sum_{n=0}^{\infty} e^{-anT} f(nT) \cdot z^{-n} \\ &= \sum_{n=0}^{\infty} f(nT) (ze^{aT})^{-n} \dots \text{--- (1)} \end{aligned}$$

$$\text{But, } \mathcal{Z}[f(t)] = F(z) = \sum_{n=0}^{\infty} f(nT) z^{-n} \dots \text{--- (2)}$$

$$\begin{aligned} \text{from (1) and (2) we have, } \mathcal{Z}[e^{-at} f(t)] &= \\ &= [F(z)]_{z \rightarrow ze^{aT}} \\ &= F(ze^{aT}). \end{aligned}$$

### 2nd shifting theorem of z-transform

State and prove "second shifting theorem of z-transform"

$$\text{If } \mathcal{Z}[f(t)] = F(z), \text{ then } \mathcal{Z}[f(t+T)] = z[F(z) - f(0)]$$

Proof

By definition,

$$\mathcal{Z}[f(t+T)] = \sum_{n=0}^{\infty} f(t+T) z^{-n}$$

$$= \sum_{n=0}^{\infty} f(nT+T) z^{-n}$$

$$= \sum_{n=0}^{\infty} f[(n+1)T] z^{-n}$$



$$\text{put } n+1 \leq N \Rightarrow n \leq N-1$$

$$\text{when } n \geq 0, N \geq 1$$

$$\text{when } n \geq 0, N \geq \infty$$

$$\therefore Z[F(t+T)] = \sum_{N=1}^{\infty} F(NT) z^{-(N-1)}$$

$$= \sum_{N=1}^{\infty} F(NT) \cdot z^{-N}$$

$$= z \left[ \sum_{N=0}^{\infty} F(NT) z^{-N} - F(0) \right]$$

$$= z [f(z) - F(0)]$$

↓  
By definition. proved

### Convolution theorem of z-transform

"Define convolution of two functions" state and prove the convolution theorem of z-transform

convolution: Let  $F(t)$  and  $g(t)$  be two functions<sup>of time</sup>. then the convolution of  $F(t)$  and  $g(t)$  is denoted by  $F(t) * g(t)$  and is defined by  $F(t) * g(t) = \sum_{k=0}^{\infty} f(kT) \cdot g(t-kT)$

### z-transform of convolution of two functions

Let  $f(z)$  and  $g(z)$  are z-transforms of  $f(t)$  and  $g(t)$  resp.

Then,  $Z[F(t) * g(t)] = F(z) \cdot g(z)$

OR "convolution of time domain property of z-transform states that the z-transform of convolution of two discrete time sequence is equal to the product of their z-transforms."



By definition  $F(z) = \sum_{k=0}^{\infty} F(kT) z^{-k}$  and

$$V(z) = \sum_{m=0}^{\infty} g(mT) z^{-m}$$

$$\downarrow$$

$$z[g(t)]$$

$$\text{Then, } F(z) \cdot V(z) = \left[ \sum_{k=0}^{\infty} F(kT) z^{-k} \right] \left[ \sum_{m=0}^{\infty} g(mT) z^{-m} \right]$$

$$= [F(0) + F(1T)z^{-1} + F(2T)z^{-2} + \dots] \left[ \sum_{m=0}^{\infty} g(mT) z^{-m} \right]$$

$$= \left[ F(0) \cdot \sum_{m=0}^{\infty} g(mT) z^{-m} + \sum_{m=0}^{\infty} g(mT) z^{-m-1} F(1T) + \sum_{m=0}^{\infty} g(mT) z^{-m-2} F(2T) + \dots \right]$$

$$= \sum_{m=0}^{\infty} F(0) g(mT) z^{-m} + \sum_{m=1}^{\infty} F(1T) g(m-1)T \cdot z^{-m} + \sum_{m=2}^{\infty} F(2T) g(m-2)T \cdot z^{-m} + \dots$$

$$= \sum_{m=0}^{\infty} [F(0) g(mT) + F(1T) \cdot g(m-1)T + F(2T) g(m-2)T + \dots] z^{-m}$$

as  $g(t)$

$g(kT)$

$g(0T)$  are zero.

$$\therefore F(z) \cdot V(z) = \sum_{m=0}^{\infty} [F(t) * g(t)] z^{-m}$$

$$= z [F(t) * g(t)]$$

proved

$$\{x: 9.1\}$$

i)

$$z[e^n]$$

$$= \sum_{n=0}^{\infty} e^n \cdot z^{-n}$$

$$= \sum_{n=0}^{\infty} \left( \frac{e}{z} \right)^n = 1 + \frac{e}{z} + \left( \frac{e}{z} \right)^2 + \left( \frac{e}{z} \right)^3 + \dots$$

iv)

$$z[r^n \cos n\theta] = ?$$

$$z[r^n \sin n\theta] = ?$$

soln

we have,  $z[a^n] = \frac{z}{z-a}$

then,  $z[(re^{i\theta})^n] = \frac{z}{1-re^{i\theta}} = \frac{z}{z-r(\cos\theta + i\sin\theta)}$

$$= \frac{z}{z-r(\cos\theta + i\sin\theta)} \times \frac{z+r(\cos\theta + i\sin\theta)}{z+r(\cos\theta + i\sin\theta)}$$

$$= \frac{z[(z+r\cos\theta) + ir\sin\theta]}{(z-r\cos\theta)^2 + r^2\sin^2\theta}$$

$$= \frac{z}{z-r(\cos\theta + i\sin\theta)} = \frac{z}{z-r\cos\theta - ir\sin\theta}$$

$$= \frac{z}{(z-r\cos\theta) - ir\sin\theta} \times \frac{(z-r\cos\theta) + ir\sin\theta}{(z-r\cos\theta) + ir\sin\theta}$$



$$= \frac{z(z - r \cos \theta) + i r z \sin \theta}{(z - r \cos \theta)^2 + r^2 \sin^2 \theta}$$

$$= \frac{z(z - r \cos \theta)}{z^2 + 2rz \cos \theta + r^2} + i \frac{r z \sin \theta}{z^2 + 2rz \cos \theta + r^2}$$

Now

$$z[r^n \cos n\theta + i r^n \sin n\theta] = \frac{z(z - r \cos \theta)}{z^2 + 2rz \cos \theta + r^2} + i \frac{r z \sin \theta}{z^2 + 2rz \cos \theta + r^2}$$

comp. the real & Imaginary part.

$$z[r^n \cos n\theta] = \frac{z(z - r \cos \theta)}{z^2 + 2rz \cos \theta + r^2}$$

$$\text{and, } z[r^n \sin n\theta] = \frac{r z \sin \theta}{z^2 + 2rz \cos \theta + r^2}$$

Initial value theorem of z-transform

If  $z[f(t)] = F(z)$ . Then  $\lim_{t \rightarrow 0} f(t) = \lim_{z \rightarrow \infty} F(z)$

That is  $\lim_{t \rightarrow \infty} F(z) = f(0)$

$$[t = nT]$$

Proof By definition  $z[f(t)] = \sum_{n=0}^{\infty} f(nT) z^{-n}$

$$\text{or, } F(z) = f(0) + F(1T)z^{-1} + F(2T)z^{-2} + \dots$$

Taking limit  $z \rightarrow \infty$  on both sides.

$$\therefore \lim_{x \rightarrow \infty} F(z) = f(0) = \lim_{t \rightarrow 0} f(t)$$



## \* final value theorem of z-transform

If  $z[f(t)] = F(z)$ . Then  $\lim_{t \rightarrow \infty} f(t) = \lim_{z \rightarrow 1} (z-1) F(z)$

Proof

By definition,

$$z[f(t)] = \sum_{n=0}^{\infty} f(nT) z^{-n}$$

Also we have by shifting theorem,

$$z[f(t+T)] = z[F(z) - f(0)]$$

$$\therefore z F(z) - z f(0) - f(z) = z[f(t+T)] - f(z)$$

$$\text{or, } (z-1) F(z) - z f(0) = \sum_{n=0}^{\infty} [f(nT+T) - f(nT)] z^{-n}$$

Taking  $\lim_{z \rightarrow 1}$  on both sides,

$$\lim_{z \rightarrow 1} [(z-1) F(z) - z f(0)] = \sum_{n=0}^{\infty} [f(nT+T) - f(nT)]$$

$$= \lim_{n \rightarrow \infty} f(nT)$$

$$= \lim_{n \rightarrow \infty} [f(T) - f(0) + f(2T) - f(T) + \dots + f(nT) - f((n-1)T)]$$

$$\therefore \lim_{z \rightarrow 1} [(z-1) F(z) - z f(0)] = \lim_{n \rightarrow \infty} f(nT) - f(0)$$

$$\text{or } \lim_{z \rightarrow 1} (z-1) F(z) - \cancel{f(0)} = f(\infty) - \cancel{f(0)}$$

$$\therefore \lim_{z \rightarrow 1} (z-1) F(z) = \lim_{t \rightarrow \infty} f(t)$$



## Shifting theorem

If  $f(n) = 0$  for  $n < 0$  such that  $z[F(t)] = f(z)$   
Then prove that:  $z[F(t-kT)] = z^{-k} f(z)$  for  $n, k > 0$

proof By definition,  
 $z[F(t)] = f(z) = \sum_{n=0}^{\infty} f(nT) z^{-n} \dots (1)$

$$\begin{aligned}\text{Then, } z[F(t-kT)] &= \sum_{n=0}^{\infty} f(nT-kT) z^{-n} \\ &= \sum_{n=0}^{\infty} f(n-k) T \cdot z^{-n}\end{aligned}$$

$$\text{put } n-k = N \Rightarrow n = k+N$$

$$\text{as } n \geq 0, N \geq -k$$

$$\text{as } n \geq \infty, N \geq \infty$$

$$z[F(t-kT)] = \sum_{N=-k}^{\infty} f(NT) z^{-k-N}$$

$$= z^{-k} \left[ \sum_{N=-k}^{\infty} f(NT) \cdot z^{-N} \right]$$

$$= z^{-k} \left[ \sum_{N=0}^{\infty} f(NT) \cdot z^{-N} \right] \text{ as } f(n) = 0 \text{ for } n < 0$$

$$\therefore z[F(t-kT)] = z^{-k} f(z) \quad \text{for } n, k > 0$$

## Theorem

If  $f(t) = 0$  for  $t < 0$  and  $z[F(t)] = f(z)$  for  $t > 0$   
Then  $z[F(t+kT)] = z^{-k} \left[ f(z) - \sum_{n=0}^{k-1} f(nT) z^{-n} \right]$

proof we have,  $f(z) = z[F(t)] = \sum_{n=0}^{\infty} f(nT) z^{-n}$

$$\text{Then, } z[F(nT+kT)] = \sum_{n=0}^{\infty} f(n+k) T \cdot z^{-n}$$



$$\text{Let, } n+k = N \Rightarrow n = N-k$$

$$\left. \begin{array}{l} \text{as } n \geq 0, N \geq k \\ \text{as } n \geq \infty, N \geq \infty \end{array} \right\}$$

$$\therefore z[F(t+kT)] = \sum_{N=k}^{\infty} f(NT) z^{-N+k}$$

$$= z^k \left[ \sum_{N=k}^{\infty} f(NT) z^{-N} \right]$$

$$= z^k \left[ \sum_{N=0}^{\infty} f(NT) z^{-N} - \sum_{N=0}^{k-1} f(NT) \cdot z^{-N} \right]$$

$$= z^k \left[ f(z) - \sum_{n=0}^{k-1} f(nT) z^{-n} \right] \therefore n \rightarrow N$$

Ex: 9.1

1) w) find z-transform of  $\cos 4\pi n$  and  $\sin 4\pi n$ .

$$\therefore z\{1\} = \frac{z}{z-1}$$

Formula, we have  $z[e^{at} f(t)] = f(z) z \rightarrow ze^{at}$

$$z[e^{-i4\pi n}] = z[f(t)] z \rightarrow ze^{4\pi i} \quad \& \quad f(t) = 1$$

$$= \left[ \frac{z}{z-1} \right] z \rightarrow ze^{4\pi i}$$

$$\frac{z \cdot ze^{4\pi i}}{ze^{4\pi i} - 1} = \frac{z}{z - e^{-4\pi i}}$$

$$= \frac{z}{z - \cos 4\pi + i \sin 4\pi}$$

$$= \frac{z}{(z - \cos 4\pi) + i \sin 4\pi} \times \frac{(z - \cos 4\pi) - i \sin 4\pi}{(z - \cos 4\pi) - i \sin 4\pi}$$



$$= \frac{z [(z - \cos 4\pi) - p \sin 4\pi]}{(z - \cos 4\pi)^2 - 1^2 \sin^2 4\pi}$$

$$= \frac{z [(z - \cos 4\pi) - p \sin 4\pi]}{z^2 - 2z \cos 4\pi + \cos^2 4\pi + \sin^2 4\pi}$$

$$= \frac{z [(z - \cos 4\pi) - p \sin 4\pi]}{z^2 - 2z \cos 4\pi + 1}$$

$$\therefore z [\cos 4\pi n + p \sin 4\pi n] = \frac{z [(z - \cos 4\pi) - p \sin 4\pi]}{z^2 - 2z \cos 4\pi + 1}$$

Equating real & imaginary

$$\therefore z [\cos 4\pi n] = \frac{z (z - \cos 4\pi)}{z^2 - 2z \cos 4\pi + 1}$$

$$\therefore z [\sin 4\pi n] = \frac{-\sin 4\pi n}{z^2 + 2z \cos 4\pi + 1}$$

Ex: 9.1

2. xQ Find z-transform of  $f(x) = \cosh 7n$

we have,  $\cosh 7n = \frac{e^{7n} + e^{-7n}}{2}$   $z[1^n] = \frac{z}{z-1}$

$$= \frac{1}{2} [(e^7)^n + (e^{-7})^n]$$

$$z [\cosh 7n] = \frac{1}{2} [z [(e^7)^n] + z [(e^{-7})^n]]$$

$$= \frac{1}{2} \left[ \frac{z}{z - e^7} + \frac{z}{z - e^{-7}} \right]$$

$$= \frac{z}{2} \left[ \frac{z - e^{-7} + z - e^7}{(z - e^7)(z - e^{-7})} \right]$$

$$= \frac{z}{2} \left[ \frac{2z - 2 \left( \frac{e^7 + e^{-7}}{2} \right)}{z^2 - ze^7 - ze^{-7} + 1} \right]$$



$$21. \therefore z[\cosh 7T] = z/2 \frac{(2z - 2\cosh 7T)}{z^2 - 2z\cosh 7T + 1}$$

2. XVIII) Find z-transform of  $(1-bt)e^{-bt}$

Soln

We have,

$$z[e^{-at}f(t)] = z[f(t)]z \rightarrow ze^{at}$$

$$z[t \cdot f(t)] = -Tz \cdot \frac{d}{dz}(z \cdot f(t)) \text{ \& } z[1] = \frac{z}{z-1}$$

$$\therefore z[(1-bt)e^{-bt}] = z[1-bt]z \rightarrow ze^{bT}$$

$$= z[1] - bTz \{f(t)\}z \rightarrow ze^{bT}$$

$$= \frac{z}{z-1} - bTz[t]z \rightarrow ze^{bT}$$

$$= \frac{z}{z-1} + bTz \frac{d}{dz}\left(\frac{z}{z-1}\right)z \rightarrow ze^{bT}$$

$$= \frac{z}{z-1} + bTz \left[ \frac{-1}{(z-1)^2} \right]z \rightarrow ze^{bT}$$

$$= \frac{z}{z-1} - bTz \cdot \frac{1}{(ze^{bT}-1)^2}$$

$$= \frac{z}{z-1} - \frac{bTz}{(ze^{bT}-1)^2}$$



## Inverse z-transform

Example

$$z[1] = \frac{z}{z-1}$$

$$z^{-1}\left[\frac{z}{z-1}\right] = 1$$

$$2) \quad z[a^n] = \frac{z}{z-a}$$

$$\therefore z^{-1}\left[\frac{z}{z-a}\right] = a^n$$

Three ways to find inverse of z transform.

① Direct division.

$$z[f(t)] = \sum_{n=0}^{\infty} f(nT) z^{-n} = f(0) + f(1) \frac{1}{z} + f(2) \frac{1}{z^2} + \dots$$

② Partial fraction method.

③ Residue technique.

z-inverse is the sum of residue of  $z^{n-1} f(z)$   
i.e.  $f(nT) = z^{-1}[f(z)] = \text{sum of residues of } z^{n-1} f(z) \text{ at poles.}$

2) pf) Find inverse z-transform of

$$f(z) = \frac{z^2 - 3z}{(z-5)(z+2)}$$

Multiplying by  $z^{n-1}$  on both side.

$$z^{n-1} f(z) = \frac{z^{n+1} - 3z^n}{(z-5)(z+2)}$$

$\therefore$  poles of  $z^{n-1} f(z)$  are  $z=5$  &  $z=-2$   
(both simple)



$$\text{then } \operatorname{Res}_{z=5} [z^{n-1} f(z)] = \lim_{z \rightarrow 5} (z-5) \left[ \frac{z^{n+1} - 3z^n}{(z-5)(z+2)} \right]$$

$$= \frac{5^{n+1} - 3 \cdot 5^n}{5+2} = \frac{5^n (5-3)}{7}$$

$$= 5^n \cdot \frac{2}{7} \sum$$

$$\text{then, } \operatorname{Res}_{z=-2} [z^{n-1} f(z)] = \lim_{z \rightarrow -2} (z+2) \left[ \frac{z^{n+1} - 3z^n}{(z-5)(z+2)} \right]$$

$$= \frac{(-2)^{n+1} - 3(-2)^n}{-2-5}$$

$$= \frac{(-2)^n (-2-3)}{-7} = (-2)^n \frac{5}{7} \sum$$

$$\text{The } z^{-1} \left[ \frac{z^2 - 3z}{(z-5)(z+2)} \right] = \text{sum of residues of } z^{n-1} f(z)$$

$$= 5^n \cdot \frac{2}{7} + (-2)^n \frac{5}{7} \sum$$

$$= \text{Ans}$$



g) Find inverse z-transform of

$$\text{Let } F(z) = \frac{z^2 - 4z}{(z-2)^2}$$

multiplying both sides by  $z^{n-1}$

$$z^{n-1} F(z) = \frac{z^{n+1} - 4z^n}{(z-2)^2}$$

clearly the pole of  $z^{n-1} F(z)$  is at  $z=2$   
(2nd order pole)

$$\therefore \text{Res} [z^{n-1} F(z)]$$

$$= \frac{1}{(n-1)!} \lim_{z \rightarrow z_0} \left[ \frac{d}{dz} (z-z_0)^n F(z) \right]$$

$$= \frac{1}{(2-1)!} \lim_{z \rightarrow 2} \left[ \frac{d}{dz} (z-2)^2 \cdot \frac{z^{n+1} - 4z^n}{(z-2)^2} \right]$$

$$= 1 \cdot \lim_{z \rightarrow 2} [(n+1)z^n - 4nz^{n-1}] = (n+1)2^n - 4n \cdot 2^{n-1}$$

$$= 2^n [n+1 - 4n \cdot \frac{1}{2}]$$

$$= 2^n [1-n]$$

$$z^{-1} \left[ \frac{z^2 - 4z}{(z-2)^2} \right] = 2^n (1-n)$$

Q



xv) Find inv. z-transform of

$$\frac{2z^3 + z}{(z-2)^2(z-1)}$$

Ans:  $9n \cdot 2^{n-1} - 2^n + 3$

### \* Application of z-transforms

z-transform can mainly be used in solving difference equations.

Difference equation: It is a relation bet<sup>n</sup> the value  $y_i$  of a function define on a discrete/lagged periodic set of argument  $x$ . It describes the patterns of change of  $y$  bet<sup>n</sup> two or more consecutive periods.

Example:  $y_{x+1} - y_x = 3$  first order diff. equation.

Application: The given diff eq<sup>n</sup> can be converted into  $\bar{y} = \psi(z)$  by taking ~~the~~ z-transform on it provided the initial values of  $y$  are known using inversion, we can get value ~~the~~  $y_x$  which is the solution of given diff. eq<sup>n</sup>

### Theorem

$$z(y_{n+k}) = z^k \left[ \bar{y} - y_0 - \frac{y_1}{z} - \frac{y_2}{z^2} - \dots - \frac{y_{k-1}}{z^{k-1}} \right]$$

proof: By definition  $z(y_{n+k}) = \sum_{n=0}^{\infty} y_{n+k} z^{-n}$

### Application of z-transforms

$$\begin{aligned} \text{or, } z(y_{n+k}) &= \sum_{n=0}^{\infty} y_{n+k} \cdot z^{-(n+k)} \cdot z^k \\ &= z^k \left[ \sum_{n=0}^{\infty} y_{n+k} z^{-(n+k)} \right] \end{aligned}$$



$$\text{put } n+k = m \Rightarrow$$

$$\text{when, } n=0, k=m$$

$$\text{when, } n=\infty, k=\infty$$

$$z(y_{n+k}) = z^k \left[ \sum_{m=k}^{\infty} y_m z^{-m} \right]$$

$$= z^k \left[ \sum_{m=0}^{\infty} y_m z^{-m} - \sum_{m=0}^{k-1} y_m z^{-m} \right]$$

$$= z^k \left[ \bar{y} - y_0 - \frac{y_1}{z} - \frac{y_2}{z^2} - \dots - \frac{y_{k-1}}{z^{k-1}} \right]$$

$$\text{put } k=1, z(y_{n+1}) = z(\bar{y} - y_0)$$

$$\text{put } k=2, z(y_{n+2}) = z^2 \left( \bar{y} - y_0 - \frac{y_1}{z} \right)$$

Ex: 9.2 solve the difference eqn.

$$9) y_{n+2} + 2y_{n+1} + y_n = 1 \text{ with } y_0 = y_1 = 0.$$

$$\text{here } y_{n+2} + 2y_{n+1} + y_n = 1 \quad \text{--- (1)}$$

$$\left. \begin{array}{l} y_0 = 0 \\ y_1 = 0 \end{array} \right\} \text{--- (2)}$$

Taking z-transform on both sides of (1).

$$z(y_{n+2}) + 2z(y_{n+1}) + z(y_n) = z[1]$$

$$\text{or, } z^2 \left( \bar{y} - \frac{y_0}{z} - \frac{y_1}{z^2} \right) + 2z \left( \bar{y} - \frac{y_0}{z} \right) + \bar{y} = \frac{z}{z-1}$$

or, using initial condition on it

$$\hookrightarrow z^2(\bar{y} - 0 - 0) + 2z(\bar{y} - 0) + \bar{y} = \frac{z}{z-1}$$

$$\hookrightarrow z^2 \bar{y} + 2z \bar{y} + \bar{y} = \frac{z}{z-1}$$



$$\text{or, } \sqrt{(z+1)^2} = \frac{z}{z-1}$$

$$\sqrt{\phantom{x}} = \frac{z}{(z-1)(z+1)^2} \quad \text{--- (1)}$$

$$z[Y_n] = \frac{z}{(z-1)(z+1)^2}$$

we need to find  $z^{-1}$  of this to get  $Y_n$

$$\text{Let } F(z) = \frac{z}{(z-1)(z+1)^2}$$

~~and~~ multiplying on both sides by  $z^{n-1}$

$$z^{n-1} F(z) = \frac{z^n}{(z-1)(z+1)^2}$$

clearly the function has poles  $z=1$  (simple pole)  
 $z=-1$  (2nd order pole)

$$\begin{aligned} \text{Res}_{z=1} [z^{n-1} F(z)] &= \lim_{z \rightarrow 1} [(z-1) z^{n-1} F(z)] \\ &= \lim_{z \rightarrow 1} (z-1) \cdot \frac{z^n}{(z-1)(z+1)^2} \\ &= \frac{1}{(1+1)^2} = \frac{1}{4} \end{aligned}$$

$$\begin{aligned} \text{Res}_{z=-1} [z^{n-1} F(z)] &= \frac{1}{(n-1)!} \cdot \lim_{z \rightarrow -1} \frac{d}{dz} [(z+1)^2 \cdot z^{n-1} F(z)] \\ &= 2 \cdot \lim_{z \rightarrow -1} \frac{d}{dz} \left[ \frac{z^n}{z-1} \right] \\ &= \frac{(z-1) \cdot n \cdot z^{n-1} - z^n}{(z-1)^2} \bigg|_{z \rightarrow -1} = \frac{1}{4} \end{aligned}$$



$$y_n = z^{-1}(\bar{y}) = \text{sum of residues of } z^{n-1} f(z) \text{ at its poles.}$$

$$= \frac{1}{4} + \frac{1}{4}$$

$$= 0$$

$$= 1 \cdot \lim_{z \rightarrow -1} \frac{(z+1)^n z^{n-1} - z^n \cdot 1}{(z+1)^2}$$

$$= \frac{(-2)^n (-1)^{n-1} - 1}{4}$$

$$y_n = z^{-1}(\bar{y}) = \text{sum of residues of } z^{n-1} f(z) \text{ at its poles,}$$

$$= \frac{1}{4} + \frac{(-2)^n (-1)^{n-1}}{4} = \frac{1 - 2^n (-1)^{n-1}}{4}$$

8)

$$y_{n+2} + 2y_{n+1} + y_n = n \quad \text{with } y_0 = y_1 = 0$$

$$\bar{z} = \frac{z}{(z+1)^2(z-1)^2} \quad \therefore z[n] = \frac{z}{(z-1)^2}$$

$$\text{res. } [z^{n-1} \cdot f(z)] = \frac{1}{(2-1)} \lim_{z \rightarrow 1} \frac{d}{dz} \left[ \frac{z^n}{(z+1)^2} \right]$$

$$= 1 \cdot \lim_{z \rightarrow 1} \left[ \frac{(z+1)^2 \cdot n z^{n-1} - z^n \cdot 2(z+1)}{(z+1)^4} \right]$$

$$= \frac{4 \cdot n - 4}{4} = \frac{n-1}{4}$$



similarly, let  $z^{n-1} f(z) = \frac{1}{(z-1)^4} \lim_{z \rightarrow -1} \frac{d}{dz} \left[ \frac{z^n}{(z-1)^2} \right]$

$$= \frac{1}{(z-1)^4} \lim_{z \rightarrow -1} \left[ \frac{(z-1)^2 + n \cdot z^{n-1} \cdot z^n \cdot 2(z-1)}{(z-1)^4} \right]$$

$$= \frac{4n(-1)^{n-1} + (-1)^n \cdot 4}{16}$$

$$= \frac{n(-1)^{n-1} + (-1)^n}{4}$$

⑧

$$4n = \frac{n-1}{4} + \frac{n(-1)^{n-1} + (-1)^n}{4}$$

$$= \frac{n-1 + n(-1)^{n-1} + (-1)^n}{4}$$